

PARIKSHA365 — STUDY NOTES

AAO — General Studies (Finance & Economics) Mastery Book

SSC CGL Tier-2 Paper III — Assistant Audit Officer / Assistant Accounts Officer
— 95%+ targeted

SSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

How to use this book

This is the only book an AAO / AAO aspirant should need.

The four rules below run through every chapter:

Rule	What it means in practice
Examiner mindset	For every topic we state how SSC frames it, the 3-5 angles they recycle, and the trap option they always plant.
Concept → Fact card → Drill	Every concept has a one-liner explanation; every section has a fact card; every chapter has 10-15 worked PYQ-style problems.
Pedagogy first	Tables, mind maps, callout boxes, depth tables. No narrative paragraphs — information is laid out like a publisher's textbook.
Self-check	Every chapter ends with a mini-mock (10 Qs, fully solved) and active-recall prompts. Last chapter is a full 100-Q mock.

***The promise.** Read this book end-to-end with two passes + the embedded recall prompts and you will solve >95 of the 100 questions on AAO Paper III.*

Paper map and importance dashboard

Paper: SSC CGL Tier-2 Paper III — **General Studies (Finance & Economics)** — for the post of **Assistant Audit Officer (AAO) / Assistant Accounts Officer**

Item	Spec
Total questions	100
Total marks	200
Marks per question	2
Negative marking	−0.5 per wrong attempt
Time	120 minutes (2 hours)
Mode	Computer-Based (CBT)
Calculator	Yes — built-in calculator allowed
Medium	English / Hindi

Paper composition

Part	Topic	Marks	Qs
A	Finance and Accounts	80	40
B	Economics and Governance	120	60

Chapter-wise importance (averaged across last 4 AAO papers)

Part A — Finance & Accounts (40 Qs / 80 marks)

Ch	Topic	Avg Qs	Importance	Difficulty
1	Fundamentals of Accounting + GAAP	5	★★★★★ HIGH	Easy
2	Double Entry, Journal, Ledger, Trial Balance	8	★★★★★ CRITICAL	Easy - Med
3	Bank Reconciliation, Rectification of Errors, Bills of Exchange	6	★★★★★ CRITICAL	Med
4	Final Accounts: Trading, P&L, P&L Appropriation, Balance Sheet	9	★★★★★ CRITICAL	Med
5	Capital vs Revenue, Depreciation, Inventory Valuation	7	★★★★★ CRITICAL	Med
6	Non-Profit Accounts, Self-Balancing Ledgers	5	★★★★ MEDIUM	Easy - Med

Part B — Economics & Governance (60 Qs / 120 marks)

Ch	Topic	Avg Qs	Importance	Difficulty
7	CAG of India + Finance Commission	6	★★★★★ CRITICAL	Easy
8	Basic Concepts of Economics + Micro intro	4	★★★★ MEDIUM	Easy
9	Theory of Demand & Supply (incl. Elasticity)	8	★★★★★ CRITICAL	Med
10	Theory of Production & Cost	6	★★★★★ HIGH	Med
11	Forms of Market & Price Determination	6	★★★★★ HIGH	Med
12	Indian Economy (sectors, NI, population, poverty, infrastructure)	8	★★★★★ CRITICAL	Easy - Med
13	Economic Reforms since 1991 (LPG + Disinvestment)	4	★★★★★ HIGH	Easy
14	Money & Banking (RBI, Commercial Banks, Monetary Policy)	7	★★★★★ CRITICAL	Med
15	Fiscal Policy, Budget, FRBM Act, Balance of Payments	7	★★★★★ CRITICAL	Med
16	Role of IT in Governance	4	★★★★ MEDIUM	Easy
	TOTAL	100		

What this dashboard tells you. The 9 CRITICAL chapters (2, 3, 4, 5, 7, 9, 12, 14, 15) carry **66 / 100 questions** = 132 / 200 marks (66%). Master them and you cross the cut-off easily. Add the 4 HIGH chapters and you cover **88%** of the paper. The 3 MEDIUM chapters are last priority.

Cut-off reality (recent AAO posts)

Year	UR cut-off (out of 200)	Ratio
2022	152–158	~77 %

Year	UR cut-off (out of 200)	Ratio
2021	148-154	~75 %
2019	145-150	~73 %

A 95% target = **190/200** demands ≥ 96 correct out of 100 (with ≤ 2 wrongs). Achievable only when every formula, fact, and definition is in muscle memory.

How to read this book — your real study load

Total pages: 91.

This is a specialised exam-prep book. **Every page is essential** — including the worked-example drills, mini-mock and trap-cards near the end. Skipping any section costs you marks.

Read end-to-end in order without skipping.

Index — Table of Contents (clickable)

Pages	Part / Chapter	Topic / What it covers
p5–6	PART 0	FOUNDATIONS YOU CANNOT SKIP
p7–9	Chapter 1	Fundamentals of Accounting & GAAP
p10–14	Chapter 2	Double Entry, Journal, Ledger & Trial Balance
p15–19	Chapter 3	Bank Reconciliation, Rectification of Errors & Bills of Exchange
p20–24	Chapter 4	Final Accounts: Trading, P&L, P&L Appropriation & Balance Sheet
p25–28	Chapter 5	Capital vs Revenue, Depreciation & Inventory Valuation
p29–32	Chapter 6	Non-Profit Organisations & Self-Balancing Ledgers
p33–36	Chapter 7	CAG of India + Finance Commission
p37–39	Chapter 8	Basic Concepts of Economics & Introduction to Microeconomics
p40–45	Chapter 9	Theory of Demand & Supply (incl. Elasticity, Consumer Behaviour)
p46–50	Chapter 10	Theory of Production & Cost
p51–55	Chapter 11	Forms of Market & Price Determination
p56–60	Chapter 12	Indian Economy
p61–63	Chapter 13	Economic Reforms in India (1991 onwards)
p64–69	Chapter 14	Money & Banking (RBI, Commercial Banks, Monetary Policy)
p70–75	Chapter 15	Fiscal Policy, Budget, FRBM Act & Balance of Payments
p76–91	Chapter 16	Role of Information Technology in Governance

PART 0 — FOUNDATIONS YOU CANNOT SKIP

Before any topic you must master these 4 micro-skills. None is hard. Skipping one will cost you 8–12 marks.

0.1 Accounting equation — the master identity

$$\text{Assets} = \text{Liabilities} + \text{Capital (Owner's Equity)}$$

This is the foundation of double-entry accounting. **Every** transaction maintains this equality. If a question's options don't satisfy this, that option is wrong by definition.

Side	What it lists
Assets	What the business owns (cash, stock, machinery, debtors, prepaid expenses)
Liabilities	What the business owes to outsiders (creditors, loans, outstanding expenses)
Capital	What the business owes to the owner (initial capital + profits – drawings – losses)

Expanded form:

$$\text{Assets} = \text{Liabilities} + \text{Capital} + \text{Revenues} - \text{Expenses} - \text{Drawings}$$

0.2 Debit-Credit golden rules (3 classes / 5 modern types)

Traditional (English) approach — by account class

Account class	Debit	Credit
Personal (persons, firms)	The receiver	The giver

Account class	Debit	Credit
Real (assets)	What comes in	What goes out
Nominal (expenses, incomes)	All expenses & losses	All incomes & gains

Modern (American) approach — by element

Element	Debit increases	Credit increases
Asset	✓	(Cr decreases)
Expense / Loss	✓	(Cr decreases)
Liability	(Dr decreases)	✓
Capital / Equity	(Dr decreases)	✓
Income / Revenue / Gain	(Dr decreases)	✓

REMEMBER

Mnemonic — DEAD CLIC.

Debits = Expenses + Assets + Drawings

Credits = Liabilities + Income + Capital

0.3 Economic micro-vocabulary you must know cold

Term	One-line meaning
GDP	Gross Domestic Product = total value of goods & services produced within a country's borders in a year
GNP	GDP + net factor income from abroad
NDP	GDP – depreciation
NNP at MP	NDP + net factor income from abroad
NNP at FC = National Income	NNP at MP – Indirect Taxes + Subsidies
Per capita income	National Income / Population
Real vs Nominal	Real = at constant prices (inflation removed); Nominal = at current prices
Inflation	Sustained rise in general price level (measured by CPI/WPI)
GDP deflator	Nominal GDP / Real GDP × 100
Fiscal deficit	Total Expenditure – (Revenue Receipts + Non-debt Capital Receipts); = government's borrowing
Revenue deficit	Revenue Expenditure – Revenue Receipts
Primary deficit	Fiscal deficit – Interest payments
Direct vs Indirect tax	Direct = paid by the entity on whom levied (income tax); Indirect = collected from intermediary, paid by consumer (GST)

0.4 Constitutional / institutional anchors (CAG, FC, RBI, EC, NITI Aayog)

Institution	Status	Article / Act	Tenure
CAG (Comptroller & Auditor General)	Constitutional, independent	Art. 148	6 years or 65 yrs, whichever earlier
Finance Commission	Constitutional, every 5 yrs	Art. 280	5-year fiscal recommendations
RBI (Reserve Bank of India)	Statutory body	RBI Act 1934 (estd 1 Apr 1935)	Governor: 3 yrs (extendable)
NITI Aayog	Executive (not constitutional/statutory)	Cabinet Resolution Jan 2015	Replaced Planning Commission
Election Commission	Constitutional	Art. 324	6 yrs or 65 yrs

CHAPTER 1 — Fundamentals of Accounting & GAAP

Importance: ★★★★★ HIGH (≈5 Qs / paper) **Difficulty:** Easy. Pure conceptual + factual.

1.0 — Understanding Accounting & GAAP from First Principles

QUICK RECALL

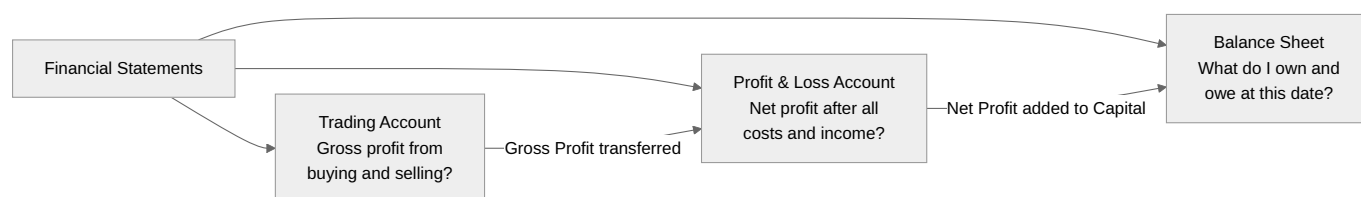
Why does accounting exist — what problem does it solve?

Imagine you start a bakery with ₹50,000 of your own money. You buy flour, sugar, an oven. You bake and sell. Three months later, you have some cash, some raw material stock, and equipment. But is the business profitable? Have all suppliers been paid? How much tax do you owe? Can you repay the bank loan?

Without a systematic record-keeping system, you cannot answer any of these reliably. **Accounting** is that system: the process of identifying, measuring, recording, classifying, summarising, interpreting, and communicating financial transactions so that all stakeholders — owner, bank, tax authority, investors — can make informed decisions.

Accounting serves four audiences simultaneously: - **Owner/Management:** Am I profitable? Where is money leaking? Should I expand? - **Banks and creditors:** Can this business repay what it owes? - **Tax authorities:** How much tax is due? - **Investors:** Is this business worth investing in?

The three core financial statements — what each answers:



GAAP — the 10 concepts that examiners test:

Concept	One-liner	Common exam question
Entity	Business is legally separate from owner	Owner's personal holiday expense → NOT a business expense
Going concern	Assume business will continue	Assets valued at cost, not liquidation price
Accrual	Record when earned/incurred, not when cash moves	Unpaid salary still an expense this period
Consistency	Same method every period	Cannot switch FIFO to LIFO year by year
Conservatism / Prudence	Anticipate losses; never anticipate profits	Write down assets to NRV if NRV < cost
Materiality	Only significant items need full disclosure	Trivial items can be approximated
Cost concept	Record at original purchase cost	Cannot inflate assets to market value
Matching	Match expenses to revenue they generate	Depreciation charged while asset earns revenue
Dual aspect	Every transaction has two effects	Foundation of double entry
Disclosure	All material facts must be disclosed	No hiding of liabilities or contingent losses

Solved Example 1.A: Owner uses company car for a personal holiday; fuel = ₹5,000. If charged as business expense, which concept is violated?

Entity concept. Business and owner are separate. Personal expense = Drawings (reduces Capital), not an operating expense.

Solved Example 1.B: Factory values machinery at current market price ₹10 lakh; historical cost was ₹4 lakh. Concept violated?

Cost concept. Assets must be recorded at historical cost. Market-value revaluation inflates assets and distorts comparability.

1.1 Examiner mindset

Angle	Pet question
Definition / scope of accounting	"Which is NOT a function of accounting?"
Accounting concepts vs conventions	Differentiate; identify the rule applied
GAAP, Accounting Standards	Body that issues AS in India
Limitations of financial accounting	Standard list of 6 limitations
Branches of accounting	Financial, Cost, Management
Users of accounting information	Internal vs external

1.2 Definition and functions

Accounting = systematic process of identifying, recording, classifying, summarizing, analyzing, interpreting, and communicating financial information about an economic entity to users.

The 4 stages (in order): 1. **Recording** → Journal (book of original entry) 2. **Classifying** → Ledger 3. **Summarizing** → Trial Balance + Final Accounts 4. **Communicating** → Reports for users

1.3 Branches of accounting

Branch	Purpose	Key output
Financial accounting	Records transactions, prepares final accounts for external users	P&L, Balance Sheet
Cost accounting	Tracks costs of products/services, enables pricing	Cost sheets
Management accounting	Provides info for internal decision-making	Budgets, MIS reports

1.4 Generally Accepted Accounting Principles (GAAP)

GAAP = the umbrella of accounting concepts + conventions + accounting standards.

Accounting Concepts (assumptions)

#	Concept	What it states
1	Business Entity	Business is separate from its owner
2	Money Measurement	Only transactions measurable in money are recorded
3	Going Concern	Business will continue indefinitely
4	Accounting Period	Life of business is divided into intervals (year/quarter)
5	Cost	Assets recorded at historical cost (not market value)
6	Dual Aspect	Every transaction has 2 aspects (Dr = Cr)
7	Realisation (Revenue Recognition)	Revenue recognised when earned, not when cash received
8	Matching	Expenses matched against revenues of the same period
9	Accrual	Record income/expense when accrued, not when cash flows
10	Objectivity	Records must be supported by verifiable evidence (vouchers)

Accounting Conventions (customs)

#	Convention	What it states
1	Conservatism (Prudence)	"Anticipate no profit, provide for all losses" — record losses early, gains late
2	Consistency	Same methods year-on-year
3	Materiality	Disclose all material (significant) facts
4	Full Disclosure	All relevant info disclosed in statements

1.5 Accounting Standards in India

Issued by	Body
Indian Accounting Standards (Ind AS)	ICAI (Institute of Chartered Accountants of India)
Notified by	Ministry of Corporate Affairs (MCA)
ICAI established	1949 under the Chartered Accountants Act, 1949
Total Ind AS notified	41 (currently in force)

1.6 Limitations of financial accounting

- Records only **monetary** transactions (ignores qualitative factors like staff morale).
- Historical** in nature — past data, may not predict future.
- Uses **estimates** (depreciation, doubtful debts) — subjective.
- Effects of **inflation** ignored (records at historical cost).
- Doesn't show **cost of products/services** (that's cost accounting).
- Window-dressing possible by manipulating disclosures.

1.7 Worked PYQ-style examples

Q 1.1. Which of the following is **not** a branch of accounting? ** (a) Financial accounting ** (b) **Cost accounting** ** (c) Management accounting ** (d) **Production accounting**

Soln. Production is an operation, not an accounting branch. **Ans (d).**

Q 1.2. "Anticipate no profit, provide for all losses" expresses which convention?

Conservatism (Prudence).

Q 1.3. Recording assets at historical cost reflects which concept?

Cost concept.

Q 1.4. ICAI was established in:

1949 (Chartered Accountants Act, 1949).

Q 1.5. GAAP stands for:

Generally Accepted Accounting Principles.

Q 1.6. A businessman records his personal car as a business asset. This violates which concept?

Business entity concept.

Q 1.7. Salary paid in advance is an example of:

Prepaid expense (an asset, not an expense, until period elapses) — reflects **matching + accrual** concepts.

Q 1.8. Recording all revenue when earned (regardless of cash collected) follows the:

Realisation / Accrual concept.

Q 1.9. Which of the following is **NOT** a limitation of financial accounting? ** (a) Ignores qualitative info ** (b) **Based on estimates** ** (c) Records all transactions ** (d) **Ignores price level changes**

(c) is wrong as a limitation — financial accounting in fact records **only monetary transactions**, not all transactions. **Ans (c).**

Q 1.10. Convention requiring uniform methods year-on-year is:

Consistency.

1.8 Trap-recognition card

Trap	Defence
GAAP issued by RBI	False — issued by ICAI , notified by MCA
"Cost concept" allows market value	No — historical cost only
Conservatism vs Prudence	Same thing, two names
Convention vs Concept confusion	Concept = assumption (entity, going-concern); Convention = practice (conservatism, consistency)

1.9 Mini-mock (10 Qs)

#	Question	Answer
1	Body issuing accounting standards in India?	ICAI
2	Going-concern means?	Business continues indefinitely
3	Dual aspect leads to which book-keeping system?	Double-entry
4	Revenue is recorded when ...?	Earned (realisation)
5	Owner's withdrawal recorded in which account?	Drawings
6	Materiality is a concept or convention?	Convention
7	First step of accounting cycle?	Identifying transactions / Recording
8	Salary paid for next year is a/an ...?	Prepaid expense (asset)
9	"All revenue cash received" violates which?	Accrual
10	Cost concept ignores?	Inflation / market value

1.10 Active-recall prompts

1. Differentiate concept vs convention with one example each.
2. List the 10 accounting concepts.
3. State the four stages of the accounting cycle.
4. List 6 limitations of financial accounting.
5. Who issues Ind AS in India?

CHAPTER 2 — Double Entry, Journal, Ledger & Trial Balance

Importance: ★★★★★ CRITICAL (≈8 Qs / paper) **Difficulty:** Easy-Medium. Master the debit-credit rule and you score full.

2.0 — Understanding Double Entry from First Principles

QUICK RECALL

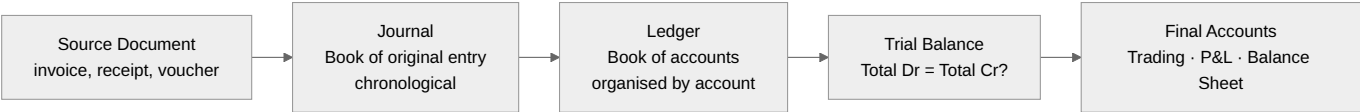
Why "double entry"? What duality does every transaction have?

Every financial transaction transfers value between two parties or accounts. When you pay ₹10,000 rent in cash: (1) your cash decreases by ₹10,000, and (2) you incur a rent expense of ₹10,000. When you sell goods worth ₹5,000 on credit: (1) your debtor's account increases by ₹5,000, and (2) your sales increase by ₹5,000. **Every transaction has exactly two economic effects.** Double entry records both — always.

This is not arbitrary bookkeeping. It reflects a fundamental economic truth: value is never created or destroyed in a transaction, only transferred. The accounting equation $\text{Assets} = \text{Liabilities} + \text{Capital}$ must hold before and after every transaction. Double entry enforces this automatically — every debit has a corresponding credit, so the equation stays in balance.

Why this makes errors visible: If a transaction is mis-recorded (one side wrong), the trial balance will not balance: total debits $\neq$ total credits. This is the built-in error detector. Any accountant who finds a mismatch in the trial balance knows immediately that at least one recording error exists.

The accounting cycle — the flow from transaction to final accounts:



Debit-Credit rules — two equivalent systems:

FORMULA

Traditional (English) system — by account type:

Account type	Debit	Credit
Personal (persons, firms, banks)	The receiver	The giver
Real (assets, stock, cash)	What comes in	What goes out
Nominal (expenses, income)	All expenses and losses	All income and gains

Modern (American) system — by account element:

Element	Debit increases	Credit increases
Asset	✓	(debit decreases)
Expense / Loss / Drawing	✓	(debit decreases)
Liability	(credit decreases)	✓
Capital / Equity	(credit decreases)	✓
Income / Revenue / Gain	(credit decreases)	✓

Mnemonic — DEAD CLIC: Debits = Expenses + Assets + Drawings; Credits = Liabilities + Income + Capital.

Solved Example 2.A — Journal entries for 5 common transactions:

#	Transaction	Journal Entry	Rule applied
1	Cash ₹50,000 introduced as capital	Cash A/c Dr 50,000 / To Capital A/c 50,000	Real (cash in) / Personal (owner = giver)
2	Goods purchased on credit from Ram, ₹20,000	Purchases A/c Dr 20,000 / To Ram's A/c 20,000	Real (stock in) / Personal (Ram = giver)
3	Cash sales ₹8,000	Cash A/c Dr 8,000 / To Sales A/c 8,000	Real (cash in) / Nominal (income = Cr)
4	Salary paid ₹5,000	Salary A/c Dr 5,000 / To Cash A/c 5,000	Nominal (expense = Dr) / Real (cash out)
5	Rent received ₹3,000	Cash A/c Dr 3,000 / To Rent Received A/c 3,000	Real (cash in) / Nominal (income = Cr)

Solved Example 2.B — Trial Balance: which errors does it catch vs. miss?

Error type	Trial Balance detects?	Why
Wrong amount on ONE side only	✓ Yes	Dr $\neq$ Cr
Error of omission (transaction not recorded at all)	✗ No	Both sides missing equally
Error of commission (wrong account, same amount)	✗ No	Dr and Cr still equal

Error type	Trial Balance detects?	Why
Error of principle (capital treated as revenue)	✗ No	Dr and Cr still equal
Compensating errors (two errors cancel each other)	✗ No	Net effect on balance = zero

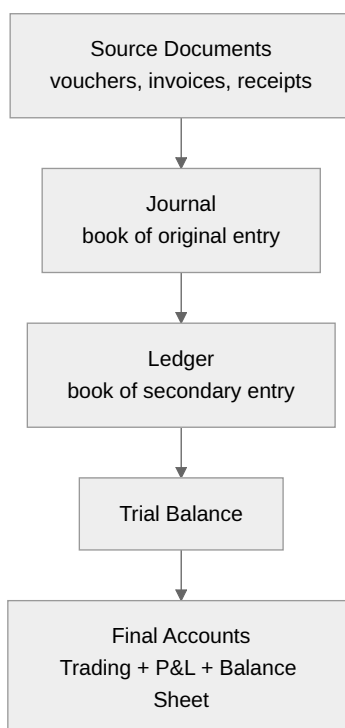
2.1 Examiner mindset

Angle	Pet question
Identify the journal entry	"Cash sales worth ₹5000 → Dr / Cr ?"
Classification of accounts	Personal / Real / Nominal
Trial Balance — purpose, errors caught vs not	"Which error is not disclosed by trial balance?"
Subsidiary books	When to use which
Suspense account	Used in rectification
Posting from journal to ledger	Direction of posting

2.2 Single Entry vs Double Entry

Aspect	Single Entry	Double Entry
Records	Only personal accounts (and cash)	Both aspects of every transaction
Trial balance	Cannot prepare	Can prepare
Profit determination	By comparison of capital	By Profit & Loss A/c
Used by	Sole traders, very small businesses	All companies, all formal accounting
Reliability	Low	High

2.3 Books of Account hierarchy



2.4 Journal entry format

Date	Particulars	L.F.	Debit (₹)	Credit (₹)
1-Apr	Cash A/c Dr.		10,000	
	To Capital A/c			10,000
	(Being capital introduced)			

2.5 Subsidiary books — when each is used

Book	Records	Posted to ledger as
Cash Book	All cash + bank receipts and payments	both journal & ledger combined
Purchases Book	Credit purchases of goods only	Total to Purchases A/c (Dr)
Sales Book	Credit sales of goods only	Total to Sales A/c (Cr)
Purchases Returns Book	Goods returned to suppliers	To Purchases Returns (Cr)
Sales Returns Book	Goods returned by customers	To Sales Returns (Dr)
Bills Receivable Book	Bills accepted in our favour	To B/R A/c (Dr)
Bills Payable Book	Bills accepted by us	To B/P A/c (Cr)
Journal Proper	Anything not in the above (opening entries, adjustments, closing entries)	direct to ledger

2.6 Cash Book types

Type	Columns	Used for
Single column	Cash only	Tiny business
Double column	Cash + Discount	Small business with discounts
Triple column	Cash + Bank + Discount	Most businesses (today)
Petty cash book	Small recurring expenses	Office expenses

2.7 Ledger — format and balancing

Date	Particulars	J.F.	Amount (Dr)	Date	Particulars	J.F.	Amount (Cr)

Balancing rule:

If Dr side total > Cr side → "By Balance c/d" on Cr side; balance is **Debit balance**.

If Cr side total > Dr side → "To Balance c/d" on Dr side; balance is **Credit balance**.

Account	Normal balance
Asset accounts	Debit
Expense accounts	Debit
Drawings	Debit
Liability accounts	Credit
Capital	Credit
Revenue / Income	Credit

2.8 Trial Balance — purpose & limits

A **trial balance** is a statement listing the debit and credit balances of all ledger accounts to **verify the arithmetical accuracy** of posting.

Errors NOT disclosed by trial balance (very high-yield):

Error type	Example	Why undetected
Error of Omission (complete)	Transaction not recorded at all	Both Dr & Cr missing
Error of Commission (offsetting)	₹500 wrongly written as ₹500 in both Dr and Cr but in wrong account	Equal Dr & Cr
Error of Principle	Capital expenditure treated as revenue (or vice versa)	Equal Dr & Cr in wrong heads
Compensating errors	Two errors cancelling each other	Net zero impact
Errors of original entry (same wrong amount on both sides)	Sales of ₹540 entered as ₹450 in both books	Equal Dr & Cr

Errors that DO show up: wrong amount on one side only, omission of one side, posting to the wrong column, casting (totalling) errors.

2.9 Worked PYQ-style examples

Q 2.1. Cash purchases of ₹2,000 — journal entry?

Purchases A/c (Dr) 2,000 / To Cash A/c (Cr) 2,000.

Q 2.2. Salary paid ₹5,000 to clerk Ram — journal entry?

Salary A/c (Dr) 5,000 / To Cash A/c 5,000. (Salary = nominal account, expense → Debit.)

Q 2.3. Sold goods to Mohan on credit ₹10,000 — entry?

Mohan's A/c (Dr) 10,000 / To Sales A/c 10,000. (Mohan is debtor — receiver in personal account.)

Q 2.4. Goods returned by Mohan ₹500 — entry?

Sales Returns A/c (Dr) 500 / To Mohan's A/c 500.

Q 2.5. Capital introduced ₹50,000 cash — entry?

Cash A/c (Dr) 50,000 / To Capital A/c 50,000.

Q 2.6. A purchase of furniture for cash worth ₹8,000. Which book?

Cash Book (it was a cash transaction, not credit). Furniture A/c gets debited from posting; cash book bank/cash credit reflects payment.

Q 2.7. Which error is **not disclosed** by Trial Balance? ** (a) Wrong total of cash book ** (b) **Posting on wrong side** ** (c) Error of principle ** (d) **Wrong ledger balance**

Ans (c). Error of principle (e.g., treating purchase of machinery as a purchase of goods) keeps Dr=Cr but is wrong by category.

Q 2.8. Sales returns book records:

Goods **returned by customers** to us (i.e., we are reducing our sales). Posted as Sales Returns A/c Dr.

Q 2.9. A trial balance ensures:

Arithmetical accuracy of posting (not freedom from all errors).

Q 2.10. Personal accounts include:

Persons (debtors, creditors), firms, banks. **Plus** representative personal accounts like "Outstanding Salary A/c" and "Prepaid Insurance A/c".

Q 2.11. Goods withdrawn by proprietor for personal use — entry?

Drawings A/c (Dr) / To Purchases A/c (since stock taken out reduces purchases at cost). Some texts use Goods A/c — both accepted.

Q 2.12. Discount allowed to a customer is which kind of account?

Nominal (it's an expense for us). Debited.

Q 2.13. A trial balance has Dr total ₹50,000, Cr total ₹49,500. Difference ₹500 is placed in:

Suspense A/c (on the Cr side) until the error is found.

Q 2.14. Closing stock not yet recorded appears in:

Trading A/c (Cr) and Balance Sheet (Asset side). NOT in trial balance unless adjustment given.

Q 2.15. Cash sales ₹3,000 entered in Sales Book by mistake. What's the impact?

Sales Book records only **credit sales**. Cash sales should be in cash book. Effect: Sales gets credited (correct) but Cash A/c is not debited; instead the customer's account would wrongly be debited if posting follows. Need rectification.

2.10 Trap-recognition card

Trap	Defence
All errors caught by trial balance	False — errors of principle, omission, compensation slip through
Sales book includes cash sales	No — only credit sales
Drawings is a liability	No — it's deducted from capital (or treated as a contra -asset)
Purchases account = stock account	No — Purchases is an expense; Stock (closing) is an asset
Prepaid insurance is an expense	No — it's a current asset until period elapses

2.11 Mini-mock

#	Question	Answer
1	Goods sold for cash. Books used?	Cash Book + Sales A/c (via posting)
2	Bill drawn on customer A is recorded in?	Bills Receivable Book
3	Trial balance verifies?	Arithmetical accuracy
4	Drawings appears on which side of capital A/c?	Debit (deduction)
5	Salary outstanding (not yet paid) is a?	Liability (representative personal account)
6	Bank overdraft is a?	Liability
7	Goodwill is which type of asset?	Intangible asset

#	Question	Answer
8	Subsidiary books are also called?	Books of original entry / Day books
9	An entry to record adjustments at year - end is in?	Journal Proper
10	Suspense account is used to balance?	Trial Balance temporarily

2.12 Active-recall prompts

1. State the three traditional rules of debit and credit.
2. List 5 errors not disclosed by trial balance.
3. Name 7 subsidiary books.
4. State which side closing stock appears on in Trading A/c and Balance Sheet.
5. Differentiate journal vs ledger.

CHAPTER 3 — Bank Reconciliation, Rectification of Errors & Bills of Exchange

Importance: ★★★★★ CRITICAL (≈6 Qs / paper) **Difficulty:** Medium. Mechanical once you know the direction of adjustment.

3.0 — Understanding BRS, Rectification & Bills from First Principles

QUICK RECALL

Why do cash book and pass book ever show different balances?

Your cash book records every cheque you write the moment you sign it. The bank (your pass book) only records it when the recipient deposits the cheque and the bank processes the clearance — which may take 2–5 days. In that window, your cash book shows a lower balance than the bank does (you've recorded the outgoing cheque; the bank hasn't processed it yet).

Similarly: your bank may credit interest or debit charges to your account before you notice. Your cash book won't have these until you record them manually. Both books are *correct* — they just reflect different snapshots in time and knowledge.

Bank Reconciliation Statement (BRS) is a systematic explanation of every difference between the two records. Its purpose is confirmation, not correction: if every difference can be accounted for logically, both records are correct.

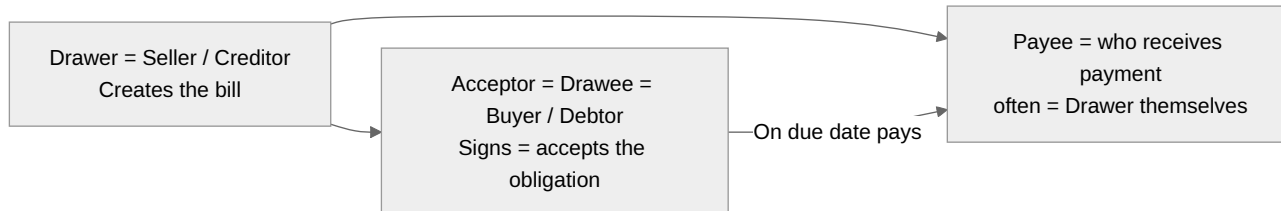
Bills of Exchange solve a different, older problem: how to formalise trade credit in a legally enforceable way. When a seller extends 90 days' credit to a buyer, the seller has an unsecured oral promise. A Bill of Exchange makes this a *written, signed, negotiable instrument* — the seller (drawer) orders the buyer (drawee) in writing to pay a specified amount on a specified date. Once the buyer signs it (acceptance), it becomes a legal obligation. The seller can then *discount* it at a bank — get cash immediately, at a small charge — rather than waiting 90 days.

BRS — the direction rule (the one thing students always confuse):

Starting from the **Cash Book balance** and adjusting to reach the **Pass Book balance**:

Situation	Effect	Adjustment to Cash Book balance
Cheque issued, not yet presented to bank	Cash Book ↓, Pass Book not yet ↓	Add back (cheque not yet cleared → PB still higher)
Cheque deposited, not yet collected by bank	Cash Book ↑, Pass Book not yet ↑	Subtract (credit in PB not yet reflected in CB)
Bank interest credited (not in CB)	Pass Book ↑, Cash Book not yet ↑	Add to CB balance
Bank charges debited (not in CB)	Pass Book ↓, Cash Book not yet ↓	Subtract from CB balance
Direct payment by bank on customer's behalf	Pass Book ↓, CB not yet ↓	Subtract from CB balance

Bills of Exchange — parties and key terms:



Term	Meaning
Drawer	The person who creates and signs the bill (the seller / creditor)
Drawee / Acceptor	The person ordered to pay; becomes the Acceptor after signing
Payee	The person who receives payment (often the drawer)
Date of maturity	Due date = date of bill + period of bill + 3 days of grace
Discounting	Selling the bill to a bank before maturity for less than face value
Endorsement	Transferring the bill to a third party by signing on the back
Dishonour	Drawee refuses or fails to pay on the due date
Noting	Official certification by a Notary Public that the bill was dishonoured

Solved Example 3.A — BRS (starting from Cash Book balance):

Cash Book balance: ₹12,000 (Dr). Find Pass Book balance given: - Cheques issued but not yet presented: ₹3,000 - Cheques deposited but not yet cleared: ₹2,000 - Bank interest credited (not in CB): ₹500 - Bank charges debited (not in CB): ₹200

*Start with CB balance: 12,000 Add back uncashed cheques: +3,000 → 15,000 Less uncollected deposits: -2,000 → 13,000 Add bank interest: +500 → 13,500 Less bank charges: -200 → **Pass Book balance = ₹13,300***

3.1 Examiner mindset

Angle	Pet question
BRS — start with cash book balance, reach pass book balance	Direct sum of adjustments
Reasons for difference	Cheques deposited but not cleared / cheques issued but not presented / direct deposits / bank charges
Rectification before / after Trial Balance	Affects suspense or directly
Bill of Exchange — three parties	Drawer / Drawee / Payee
Discounting / endorsing / retiring of bill	Standard 4 cases

3.2 Bank Reconciliation Statement (BRS)

*A statement reconciling the **cash book bank balance** with the **pass book balance** at a given date.*

Standard differences and their direction

When starting from **Cash Book balance (Dr / favourable)** and reaching **Pass Book balance (Cr / favourable)**:

Reason	Effect on cash book vs pass book	In BRS:
Cheque issued but not yet presented for payment	Pass book higher	Add
Cheque deposited but not yet cleared	Pass book lower	Subtract
Bank charges debited by bank (not in cash book)	Pass book lower	Subtract
Interest credited by bank (not in cash book)	Pass book higher	Add
Direct deposit by customer in our account	Pass book higher	Add
Direct payment by bank (standing instruction, e.g., LIC premium)	Pass book lower	Subtract
Cheque dishonoured (already credited in cash book)	Pass book lower	Subtract

Rule of thumb:

Add items that **increase pass book balance** vs cash book.

Subtract items that **decrease pass book balance** vs cash book.

(Reverse all signs if starting from pass book balance.)

Format

Particulars	₹
Balance as per Cash Book	xxx
Add: cheques issued but not presented	xxx
Add: interest credited by bank	xxx
Less: cheques deposited but not cleared	(xxx)
Less: bank charges	(xxx)
Balance as per Pass Book	xxx

3.3 Rectification of errors — three scenarios

Case A: Error found before preparation of Trial Balance

Pass a single rectifying journal entry in the original books — no suspense account.

Case B: Error found after trial balance but before final accounts

Use **Suspense A/c** to balance the trial balance, then pass rectifying entries that route through Suspense.

Case C: Error found after preparation of final accounts

Use **Profit & Loss Adjustment A/c** + Suspense.

Common rectification patterns

Original error	Rectification entry (after TB)
Sales of ₹500 omitted	Customer's A/c Dr 500 / To Sales A/c 500
Purchases A/c short by ₹200	Purchases A/c Dr 200 / To Suspense A/c 200
Salary ₹1000 posted as ₹100 in salary A/c	Salary A/c Dr 900 / To Suspense A/c 900

Original error	Rectification entry (after TB)
Cash received from Ram ₹500, posted to Shyam	Shyam Dr 500 / To Ram 500
Goods purchased on credit ₹1000 posted to wrong side	Purchases Dr 2000 / To Suspense 2000 (× 2 to undo + correct)

3.4 Bills of Exchange

*A **Bill of Exchange** is a written, unconditional order signed by the maker (drawer), directing a certain person (drawee) to pay a certain sum of money to a specified person (payee) on a specified future date.*

Term	Meaning
Drawer	Person who makes the bill (creditor)
Drawee	Person on whom bill is drawn (debtor); becomes acceptor after signing
Payee	Person to whom payment is to be made (often = drawer)
Tenure	Time between drawing and maturity
Maturity date	Due date + 3 days of grace
Discounting	Drawer takes the bill to a bank before maturity, gets cash less discount
Endorsement	Transfer of bill to another party (signing on the back)
Retiring	Settling the bill before maturity (with rebate)
Renewal	Drawing a new bill in place of old (with interest) for unable-to-pay debtor
Dishonour	Drawee refuses / fails to pay on maturity → bill returned, possibly with noting charges

Journal entries (drawer's books)

Event	Entry
Bill drawn on Mohan & accepted	Bills Receivable A/c Dr / To Mohan A/c
Bill discounted with bank	Bank A/c Dr / Discount A/c Dr / To Bills Receivable A/c
Bill endorsed to Sohan	Sohan A/c Dr / To Bills Receivable A/c
Bill honoured on due date	Bank A/c Dr / To Bills Receivable A/c
Bill dishonoured	Mohan A/c Dr / To Bills Receivable A/c (+ noting charges)

3.5 Worked PYQ-style examples

Q 3.1. Cash book balance ₹5,000 (Dr). Cheques issued not presented ₹1,200; cheques deposited not cleared ₹800. Pass book balance?

- Start: Cash Book balance = ₹5,000
- Add cheques issued not presented = +₹1,200 → ₹6,200
- Less cheques deposited not cleared = -₹800
- Pass Book balance = **₹5,400** (favourable)

Q 3.2. Pass book balance ₹3,000 (Cr). Bank charges ₹50; interest credited ₹150. Find cash book balance.

Reverse logic: starting from pass book, to find cash book, apply **opposite** signs. - Less: interest credited (add to PB but not to CB) = -150 - Add: bank charges (deducted from PB but not from CB) = +50 - Cash book balance = 3,000 - 150 + 50 = **₹2,900**

Q 3.3. A bill of ₹10,000 drawn on 5 April for 3 months. Maturity date?

- Date of drawing + tenure = 5 April + 3 months = 5 July
- Add 3 days of grace = **8 July**

Q 3.4. Cheque deposited but not yet cleared by bank. In BRS starting from cash book balance, this is:

Subtracted (because cash book has already added it but pass book hasn't yet).

Q 3.5. A salary of ₹5,000 was posted twice in the salary A/c. Rectification (after TB)?

Suspense A/c Dr 5,000 / To Salary A/c 5,000.

Q 3.6. Drawer of bill is the:

Creditor / seller (the one who is owed money and writes the bill).

Q 3.7. Bill discounted with bank ₹10,000 at 6 % for 3 months. Discount?

- = $10,000 \times 6/100 \times 3/12$
- = $10,000 \times 0.015$
- = **₹150**
- Bank credits = $10,000 - 150 = ₹9,850$

Q 3.8. A cheque of ₹1,000 issued to Ram was wrongly recorded as ₹100 in cash book. In BRS (starting from cash book balance):

Cash book is overstated (less paid recorded); to reach correct pass book → **Subtract ₹900**.

Q 3.9. Bill dishonoured — the drawer's entry?

Drawee's A/c (Dr) / To Bills Receivable A/c. Plus noting charges A/c (Dr) / To cash if any.

Q 3.10. A direct deposit by customer Ram ₹3,000 — in BRS starting from cash book?

Add ₹3,000 (pass book has it, cash book doesn't yet).

3.6 Trap-recognition card

Trap	Defence
Cheque issued but not yet presented = subtract	No — add (pass book balance is higher because money is still in account).
Bank charges added in BRS	No — subtracted (pass book is lower).
Days of grace = 5	No — 3 days .
Drawee is the creditor	No — drawer is creditor; drawee is debtor.
All errors need suspense account	Only those found after TB.

3.7 Mini-mock

#	Question	Answer
1	Days of grace on a bill	3
2	Bill drawn 1 Mar for 2 months — maturity?	4 May (1 May + 3 days)
3	Drawer of a bill is also called?	Maker
4	Discounting a bill records discount as?	Expense (Dr)
5	Cheque deposited but not cleared — pass book is?	Lower than cash book
6	Direct deposit by customer — in BRS (from cash book)?	Add
7	Bill payable book entries are by?	Drawee (acceptor)
8	Suspense A/c is used until?	Errors are located
9	Bill of exchange minimum parties?	3
10	Bill renewal involves?	Cancelling old + drawing new bill (often with interest)

3.8 Active-recall prompts

1. List 6 reasons for difference between cash book and pass book.
2. State the rule for adding/subtracting in BRS starting from cash book.
3. Define drawer, drawee, payee.
4. State maturity date formula.
5. Differentiate retiring vs renewal of bill.

CHAPTER 4 — Final Accounts: Trading, P&L, P&L Appropriation & Balance Sheet

Importance: ★★★★★ CRITICAL (≈9 Qs / paper) **Difficulty:** Medium. The most important chapter in Part A. Master adjustments and you score full.

4.0 — Understanding Final Accounts from First Principles

QUICK RECALL

What do the three financial statements actually tell different stakeholders?

Think of a business as having two questions: "How did we do this year?" and "Where do we stand today?" The three final accounts answer these in a specific sequence:

Step 1 — Trading Account: "How much money did we make from our core trade?" Take all sales → subtract the direct cost of goods sold (opening stock + purchases – closing stock) → get **Gross Profit**. This is profit before any operating overhead. A bakery's trading account shows: Sales ₹5,00,000 – Cost of goods sold ₹3,00,000 = Gross Profit ₹2,00,000. The baker hasn't yet paid salaries, rent, or electricity — that comes next.

Step 2 — Profit & Loss Account: "After all overheads and other income, what's our net profit?" Take Gross Profit → subtract indirect expenses (salaries, rent, depreciation, advertising) → add indirect income (commission received, interest received) → get **Net Profit**. Continuing the bakery: ₹2,00,000 – (Salaries ₹80,000 + Rent ₹40,000 + Misc ₹20,000) = Net Profit ₹60,000.

Step 3 — Balance Sheet: "At the end of the year, what do we own and what do we owe?" Assets on one side, Liabilities + Capital on the other. Net Profit gets added to Capital on the Balance Sheet — this is how income statement and balance sheet are linked. The Balance Sheet must balance: Assets = Liabilities + Capital.

The flow — adjustments are the examiner's tool:

Every adjustment must appear in **two places** (once in Trading/P&L and once in Balance Sheet). Missing the second entry is the single most common error.

Adjustment	Trading / P&L effect	Balance Sheet effect
Closing stock	Credit in Trading A/c (reduces cost of goods sold)	Current Asset
Outstanding expense	Debit in P&L (add to the expense)	Current Liability
Prepaid expense	Credit in P&L (deduct from the expense)	Current Asset
Accrued income	Debit in P&L (add to the income)	Current Asset
Income received in advance	Credit in P&L (deduct from income)	Current Liability
Depreciation	Debit in P&L	Reduce fixed asset value
Bad debts	Debit in P&L	Reduce debtors in BS
Provision for bad debts	Debit in P&L	Reduce debtors in BS

Direct vs indirect expenses — the line that separates Trading from P&L:

Direct expenses (go to Trading A/c)	Indirect expenses (go to P&L)
Purchases, carriage inward, wages (factory), customs duty, octroi, freight	Salaries, office rent, advertising, depreciation, bad debts, commission paid, bank charges

KEY FACT

Key distinction: "Wages" in a manufacturing context = direct (factory workers making the product). "Salaries" = indirect (office staff, management). Examiners frequently test this boundary.

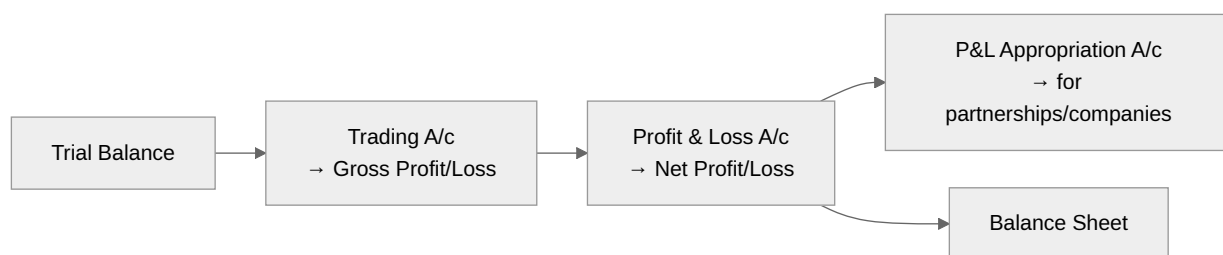
Solved Example 4.A: Closing stock ₹15,000; outstanding wages ₹3,000; prepaid insurance ₹2,000; depreciation on machinery ₹10,000. Show where each appears.

- *Closing stock: Credit in Trading A/c; Current Asset in BS (Asset side).*
- *Outstanding wages: Added to wages in Trading/P&L; Current Liability in BS.*
- *Prepaid insurance: Deducted from insurance in P&L; Current Asset in BS.*
- *Depreciation: Debit in P&L; Deducted from Machinery value in BS.*

4.1 Examiner mindset

Angle	Pet question
Items above / below the gross-profit line	Direct expenses vs indirect
Items in P&L vs Balance Sheet	Salary outstanding vs Salary paid
Adjustment entries (closing stock, accruals, prepaid, depreciation, bad debts)	One-line "two-fold effect" question
Difference between Trading and P&L	Trading → Gross profit; P&L → Net profit
Items in Capital A/c	Opening capital + addition – drawings + profit
Operating profit / EBIT / EBT	Definitions

4.2 The Final Accounts ladder



4.3 Trading Account

Purpose: Compute **Gross Profit (GP)** = Net Sales – Cost of Goods Sold.

Debit (left)	Credit (right)
Opening stock	Sales (gross) – Sales returns
Purchases (gross) – Purchases returns	Closing stock
Direct expenses: wages, freight inward, carriage inward, factory rent, fuel, power, octroi, dock charges, royalty on production	Gross Profit c/d (if Cr > Dr)
Gross Profit c/d (if Dr < Cr)	Gross Loss c/d (if Dr > Cr)

Cost of Goods Sold (COGS) formula:

$$\text{COGS} = \text{Opening Stock} + \text{Net Purchases} + \text{Direct Expenses} - \text{Closing Stock}$$

$$\text{Gross Profit} = \text{Net Sales} - \text{COGS}$$

4.4 Profit & Loss Account

Purpose: Compute **Net Profit (NP)** = Gross Profit – Indirect (operating + non-operating) Expenses + Other Incomes.

Debit (Indirect Expenses)	Credit
Office & admin: salary, rent, lighting, insurance, telephone, audit fees, legal charges	Gross Profit b/d
Selling & distribution: advertisement, commission, carriage outward, packing, salesmen salary, travelling expenses, bad debts, discount allowed	Other incomes: rent received, interest received, commission received, discount received
Financial: interest on loan, bank charges	Net Loss c/d (rare)
Depreciation, Provisions (for doubtful debts, discounts), Loss on sale of asset	
Net Profit c/d (transferred to Capital)	

4.5 Balance Sheet (vertical or horizontal)

Purpose: A statement of assets, liabilities and capital at a point in time.

Horizontal format

Liabilities	₹	Assets	₹
Capital	xxx	Fixed assets: Land & Building, Plant & Machinery, Furniture, Vehicles, Goodwill	xxx
Add: Net profit, Less: Drawings		Investments	xxx
Long-term loans (debentures, term loans)		Current assets: Closing stock, Debtors, B/R, Cash, Bank, Prepaid expenses, Accrued income	
Current liabilities: Creditors, B/P, Outstanding expenses, Income received in advance, Bank overdraft		Fictitious assets: Discount on issue of debentures, Preliminary expenses (deferred)	

Marshalling order: Either by **liquidity** (most liquid first) or by **permanence** (most permanent first). Both are acceptable but consistent within a statement.

4.6 Adjustment entries — the "twofold effect" matrix

Adjustment	Trading A/c effect	P&L A/c effect	Balance Sheet effect
Closing stock	Cr side (Trading)	—	Asset (Current)

Adjustment	Trading A/c effect	P&L A/c effect	Balance Sheet effect
Outstanding expenses (e.g., salary unpaid)	Add to that expense (Dr in P&L)	—	Current liability
Prepaid expenses	Subtract from that expense (Dr in P&L)	—	Current asset
Accrued income	—	Add to income (Cr in P&L)	Current asset
Income received in advance	—	Subtract from income (Cr in P&L)	Current liability
Depreciation	—	Dr in P&L	Subtract from asset
Bad debts (additional)	—	Dr in P&L	Subtract from Debtors
Provision for doubtful debts (new)	—	Dr in P&L	Subtract from Debtors
Provision (reversal of old, increase of new)	—	Net effect Dr in P&L	Updated provision in B/S
Interest on capital	—	Dr in P&L	Add to Capital
Interest on drawings	—	Cr in P&L	Subtract from Capital
Goods withdrawn by owner	Subtract from purchases	—	Subtract from Capital (Drawings)

REMEMBER

Universal rule. Every adjustment given **outside** the trial balance has a **two-fold** effect. If you record only one effect, your P&L and B/S will not tally.

4.7 Profit & Loss Appropriation A/c (partnerships)

*Used to allocate net profit between partners after providing for **interest on capital, interest on drawings, partner's salary, partner's commission, and reserve transfers.***

Debit	Credit
Interest on capital (to partners)	Net Profit b/d
Salary / commission to partners	Interest on drawings (from partners)
Reserve transfer	
Profit to partners' capitals (in P&L sharing ratio)	

4.8 Worked PYQ-style examples

Q 4.1. Net Sales ₹50,000; COGS ₹30,000. Gross profit?

$50,000 - 30,000 = \text{₹}20,000$.

Q 4.2. Opening stock ₹5,000; Purchases ₹40,000; Closing stock ₹3,000. COGS?

- $= 5,000 + 40,000 - 3,000$
- $= \text{₹}42,000$

Q 4.3. Salaries ₹10,000; Outstanding salary ₹2,000. Salary expense in P&L?

$10,000 + 2,000 = \text{₹}12,000$. Outstanding ₹2,000 also appears as a liability in Balance Sheet.

Q 4.4. Insurance prepaid ₹500 out of ₹3,000 paid. Insurance expense in P&L?

$3,000 - 500 = \text{₹}2,500$. Prepaid ₹500 = current asset in B/S.

Q 4.5. Depreciation 10 % p.a. on plant ₹50,000. Depreciation expense?

- Depreciation $= 50,000 \times 0.10 = \text{₹}5,000$
- Plant in B/S $= 50,000 - 5,000 = \text{₹}45,000$

Q 4.6. Carriage inward goes in:

Trading A/c (debit) — it's a direct expense of bringing goods.

Q 4.7. Carriage outward goes in:

P&L A/c (debit) — distribution expense.

Q 4.8. Goods withdrawn by owner ₹1,000. Effect on Trading A/c?

Reduces purchases → Cr Purchases A/c by ₹1,000 (or shown on Cr side of Trading). Also Drawings increases.

Q 4.9. Closing stock at cost ₹10,000, market value ₹8,000. Value taken?

Lower of cost or market = ₹8,000 (Conservatism).

Q 4.10. Bad debts ₹500 (already in TB). Further bad debts ₹200 to write off and a provision @ 5 % on net debtors needed; debtors ₹20,000.

- Net debtors after further bad debts $= 20,000 - 200 = \text{₹}19,800$
- Provision @ 5% $= 19,800 \times 5/100 = \text{₹}990$
- Total P&L Dr $= 500$ (existing) $+ 200$ (further) $+ 990$ (provision) $= \text{₹}1,690$
- Balance Sheet Debtors $= 20,000 - 200 - 990 = \text{₹}18,810$

Q 4.11. Interest on capital is debited to:

P&L Appropriation A/c (in partnerships); P&L A/c (in sole-trader, treated as expense to owner).

Q 4.12. Goodwill is which type of asset?

Intangible fixed asset.

Q 4.13. Outstanding wages of ₹500 — entry?

Wages A/c Dr 500 / To Outstanding Wages A/c 500.

Q 4.14. Net profit ₹50,000; Drawings ₹10,000; Interest on capital ₹5,000 (already adjusted in P&L). Closing capital = Opening capital + ?

- Net profit – Drawings $= +50,000 - 10,000 = +40,000$. (IoC already inside profit calc.)

Q 4.15. Sales ₹1,00,000; Opening stock ₹15,000; Purchases ₹70,000; Closing stock ₹10,000; Wages ₹5,000. Gross profit?

- COGS $= 15,000 + 70,000 + 5,000 - 10,000 = \text{₹}80,000$
- GP $= 1,00,000 - 80,000 = \text{₹}20,000$

4.9 Trap-recognition card

Trap	Defence
Carriage outward in Trading A/c	No — P&L A/c (it's a selling expense)
Returns inward = Sales returns	True. Returns outward = Purchase returns.
Closing stock in trial balance → no Cr in trading	If already in TB, only B/S. If only in adjustments, both trading (Cr) and B/S (Asset)
Drawings = expense	No — it's a reduction from capital, not an expense.
Interest on capital = expense for partnership	It's an appropriation of profit (P&L Appropriation), not P&L.

4.10 Mini-mock

#	Q	Answer
1	Direct expenses go in?	Trading A/c (Dr)
2	Closing stock valued at?	Lower of cost or market
3	Depreciation in B/S?	Reduce from fixed asset

#	Q	Answer
4	Outstanding rent in B/S?	Current liability
5	Net profit transferred to?	Capital A/c (Cr)
6	Goodwill is?	Intangible fixed asset
7	Bank overdraft in B/S?	Current liability
8	Returns outward = ?	Purchase returns
9	Prepaid insurance in B/S?	Current asset
10	Operating profit = ?	GP – Operating expenses

4.11 Active-recall prompts

1. State the COGS formula.
2. Differentiate Trading A/c vs P&L A/c.
3. List 8 direct expenses.
4. Give the two-fold effect of (a) outstanding salary, (b) accrued interest, (c) depreciation.
5. Define operating profit, EBIT, EBT.

CHAPTER 5 — Capital vs Revenue, Depreciation & Inventory Valuation

Importance: ★★★★★ CRITICAL (≈7 Qs / paper) **Difficulty:** Medium. Definition + small computation. Memorise depreciation methods and FIFO/LIFO.

5.0 — Understanding Capital vs Revenue, Depreciation & Inventory

QUICK RECALL

Why must we separate capital and revenue expenditure? What goes wrong if we don't?

A company buys a delivery truck for ₹8,00,000. This truck will serve for 8 years. If the full ₹8,00,000 is treated as an expense this year, profit is wiped out, making this year look terrible. Next 7 years the truck runs at zero "truck cost," making them look artificially profitable. Both pictures are wrong.

The correct approach: **spread the cost** over 8 years — charge ₹1,00,000 as depreciation expense each year. Each year's profit now accurately reflects that year's portion of the truck's value consumed.

The classification rule: - **Capital expenditure:** Acquires an asset with benefits lasting more than one accounting period. It appears on the Balance Sheet (as an asset, reduced by accumulated depreciation). - **Revenue expenditure:** Consumed in the current period. Charged fully to P&L in the current year.

Inventory valuation directly affects profit: The value assigned to closing stock determines Cost of Goods Sold (COGS = Opening Stock + Purchases – Closing Stock). If you overvalue closing stock, COGS falls → Gross Profit rises → you report higher profit. This is why AS-2 mandates: value at **cost or net realisable value, whichever is lower** — the prudence principle. AS-2 allows **FIFO** and **Weighted Average**, but **prohibits LIFO**.

SLM vs WDV — the intuition behind each method:

Aspect	Straight Line Method (SLM)	Written Down Value (WDV)
Depreciation amount	Same every year: $\frac{C-S}{n}$	Decreasing: fixed % of book value
Formula	Annual dep = $\frac{\text{Cost} - \text{Scrap}}{\text{Useful life}}$	Rate = $1 - \left(\frac{S}{C}\right)^{1/n}$
Asset type best suited	Assets that wear out uniformly	Assets more valuable when new (cars, computers)
Book value at end	Zero (or scrap value)	Never reaches zero
Exam trap	Sum of all depreciation = Cost – Scrap	Rate % is applied to opening book value each year

Solved Example 5.A — SLM vs WDV on the same asset:

Machine: Cost ₹1,00,000. Scrap ₹10,000. Useful life = 9 years.

SLM: Annual depreciation = $(1,00,000 - 10,000)/9 = ₹10,000$ per year. Book value falls by ₹10,000 each year.

WDV at 25%: - Year 1 depreciation = $1,00,000 \times 25\% = ₹25,000$ - Book value end of Year 1 = $1,00,000 - 25,000 = ₹75,000$ - Year 2 depreciation = $75,000 \times 25\% = ₹18,750$ - Book value end of Year 2 = $75,000 - 18,750 = ₹56,250$ - WDV charges more in early years — gives a tax benefit sooner.

Solved Example 5.B — FIFO vs Weighted Average for inventory:

Opening stock: 100 units @ ₹10 = ₹1,000. Purchased: 200 units @ ₹12 = ₹2,400. Sold: 150 units.

FIFO: - Sell 100 units (opening stock) @ ₹10 = ₹1,000 - Sell 50 more units (from purchase batch) @ ₹12 = ₹600 - COGS = $1,000 + 600 = ₹1,600$ - Closing stock = 150 units @ ₹12 = ₹1,800

Weighted Average: - Total cost = $1,000 + 2,400 = ₹3,400$ - Total units = $100 + 200 = 300$ - Average cost per unit = $3,400 / 300 = ₹11.33$ - COGS = $150 \times 11.33 = ₹1,700$ - Closing stock = $150 \times 11.33 = ₹1,700$

In rising prices: FIFO gives lower COGS → higher profit. Weighted average smooths fluctuations.

5.1 Examiner mindset

Angle	Pot question
Capital vs Revenue expenditure / receipt	Classify the transaction
Deferred revenue expenditure	"Heavy advertising whose benefit lasts 3 years"
Depreciation methods	SLM vs WDV computation
Causes of depreciation	Wear, obsolescence, time, depletion
Inventory valuation methods	FIFO, LIFO, Weighted Average, Specific
Cost or NRV (lower of)	Conservatism

5.2 Capital vs Revenue expenditure

Capital	Revenue
Benefit > 1 accounting year	Benefit within current year
Increases earning capacity / extends life of asset	Maintains existing capacity
Shown as asset in B/S	Shown as expense in P&L

Capital	Revenue
Examples: Purchase of machinery, building extension, major overhaul that extends life, legal fees on purchase of land	Examples: Repairs & maintenance, salary, electricity, raw material, ordinary wages

Deferred Revenue Expenditure

What	Heavy revenue expense whose benefit extends over multiple years
Examples	Heavy advertisement campaign, R&D for a product, preliminary expenses (now disallowed by Ind AS), loss by fire (uninsured)
Treatment	Charged to P&L over benefit-period; unwritten portion in B/S as fictitious asset

5.3 Capital vs Revenue receipt

Capital receipt	Revenue receipt
Non-recurring	Recurring
Reduces an asset or increases a liability	Earned from operations
Goes to Balance Sheet	Goes to P&L (Cr)
Examples: Capital introduced, sale of fixed asset, loan taken	Examples: Sales, interest, rent, commission, discount received

5.4 Depreciation — meaning, causes, methods

Depreciation = systematic allocation of the cost of a tangible fixed asset over its useful life.

Causes (memorise the list)

1. **Wear and tear** (physical use)
2. **Effluxion of time** (e.g., leasehold expiring)
3. **Obsolescence** (technology change)
4. **Depletion** (extractive assets like mines)
5. **Accident**

Methods of depreciation

Method	Formula	Pattern
Straight Line Method (SLM)	(Cost – Salvage) / Useful life	Equal annual amount
Written Down Value (WDV) / Diminishing balance	WDV at start × rate %	Decreasing annual amount
Sum of Years' Digits	(Remaining life / Σ years) × (Cost – Salvage)	Decreasing
Units of production	(Cost – Salvage) / Total units × units this year	Variable with output
Annuity	Considers time value of money on the investment	Increasing
Sinking fund	Sets aside fund for replacement	Equal annual amount
Insurance policy method	Premium paid each year acts as depreciation	Equal

SLM vs WDV — the only comparison asked

Feature	SLM	WDV
Annual depreciation amount	Constant	Decreasing
Asset value reaches zero	Yes (at end of useful life)	Never reaches zero (asymptotic)
Total maintenance + depreciation	Increases over life	More uniform
Ease of calculation	Simple	Slightly complex
Suitable for	Lease, low-tech assets	Plant & machinery, vehicles
Tax law in India	WDV is the income-tax method	(SLM allowed for some PSU)

WDV rate from useful life

$$r = 1 - \left(\frac{S}{C}\right)^{1/n}$$

where C = cost, S = salvage value, n = useful life. Asked as: "find rate to reduce ₹X to ₹Y in n years."

Methods of recording depreciation

1. **Asset A/c method** — Asset is directly reduced; loss to Depreciation A/c.

2. **Provision for depreciation A/c method** — Asset shown at original cost; cumulative depreciation in a separate provision account; net = Cost – Provision.

5.5 Inventory valuation methods

Method	When prices are rising	Effect on COGS / Profit
FIFO (First-In-First-Out)	Closing stock at recent (high) prices	Lower COGS, higher profit
LIFO (Last-In-First-Out)	Closing stock at older (low) prices	Higher COGS, lower profit (tax-saving)
Weighted Average	Smoothed prices	Between FIFO and LIFO
Specific identification	Track individual item cost	Used for unique items (jewels, cars)

Indian standard. AS-2 (and Ind AS 2) **prohibits LIFO**. Only FIFO and Weighted Average are allowed for inventory valuation in India. (LIFO is allowed in US GAAP.)

Cost vs Net Realisable Value (NRV)

Closing stock is valued at **Lower of Cost or Net Realisable Value** (Conservatism).

NRV = Estimated selling price – cost to complete and sell.

5.6 Worked PYQ-style examples

Q 5.1. Purchase of a delivery van — capital or revenue?

Capital (long-term asset).

Q 5.2. Repairs on the same van after 1 year — capital or revenue?

Revenue (maintains, doesn't extend life).

Q 5.3. Plant cost ₹100,000, salvage ₹10,000, useful life 9 years. SLM annual depreciation?

$(100,000 - 10,000)/9 = \text{₹}10,000$.

Q 5.4. WDV depreciation @ 20 % p.a. on plant ₹100,000. Year 1 and Year 2 depreciation?

- Year 1 depreciation = $100,000 \times 0.20 = \text{₹}20,000$
- WDV at start of Year 2 = $100,000 - 20,000 = \text{₹}80,000$
- Year 2 depreciation = $80,000 \times 0.20 = \text{₹}16,000$
- WDV at end of Year 2 = $80,000 - 16,000 = \text{₹}64,000$

Q 5.5. A heavy advertising campaign of ₹3,00,000 whose benefit lasts 3 years. P&L charge per year?

₹100,000 (deferred revenue expenditure spread over benefit period).

Q 5.6. Closing stock cost ₹50,000, NRV ₹45,000. Recorded value?

₹45,000 (lower of cost or NRV).

Q 5.7. FIFO during a period of rising prices gives:

Higher closing stock value, **lower** COGS, **higher** profit (vs LIFO).

Q 5.8. A machinery costing ₹50,000 was sold for ₹40,000 after 2 years. SLM was 10 % p.a. WDV at sale?

- Annual SLM depreciation = $50,000 \times 10\% = \text{₹}5,000$
- WDV after 2 years = $50,000 - 2 \times 5,000 = \text{₹}40,000$
- Sold at ₹40,000 = WDV → **No profit, no loss**

Q 5.9. Method of depreciation prescribed by Income Tax Act in India?

WDV (with block of assets concept).

Q 5.10. AS-2 in India prohibits which inventory valuation method?

LIFO.

Q 5.11. Goodwill is amortised under which standard?

Treated under **AS-26 / Ind AS 38** (Intangible assets).

Q 5.12. A premium paid on issue of debentures is — capital / revenue / deferred?

Capital receipt (premium on debentures = capital reserve).

Q 5.13. Building purchased ₹500,000 + legal fees ₹20,000 + brokerage ₹10,000. Cost capitalised?

₹530,000 (all costs to make asset ready for use).

Q 5.14. Plant cost ₹100,000; SLM @ 10 % for 3 years; sold at ₹80,000. Profit / loss?

- Annual SLM depreciation = $100,000 \times 10\% = \text{₹}10,000$
- WDV after 3 years = $100,000 - 3 \times 10,000 = \text{₹}70,000$
- Profit on sale = $80,000 - 70,000 = \text{₹}10,000$ (capital profit on sale of fixed asset)

5.7 Trap-recognition card

Trap	Defence
Repairs that increase capacity = revenue	No — capital (extends/improves asset)
Major overhaul = revenue	If it extends life materially → capital
WDV asset reaches zero	No — asymptotic
LIFO valid in India	No — prohibited by AS-2
Closing stock at cost always	No — lower of cost or NRV
Capital receipt in P&L	No — Balance Sheet

5.8 Mini-mock

#	Q	Answer
1	SLM annual amount = ?	(Cost – Salvage)/Life
2	Method allowed under Indian Income Tax?	WDV (block of assets)
3	Conservatism applied to inventory?	Lower of cost or NRV
4	Sale of fixed asset is which receipt?	Capital
5	Heavy advertisement spread over years is?	Deferred revenue expenditure
6	FIFO in rising prices → profit is?	Higher
7	Depreciation is a — cash / non-cash expense?	Non-cash
8	A leasehold building's depreciation reflects?	Effluxion of time
9	Mining asset depreciated by?	Depletion
10	Specific identification suitable for?	Unique high-value items (cars, jewels)

5.9 Active-recall prompts

1. List 5 causes of depreciation.
2. Compare SLM and WDV on 3 dimensions.
3. State which inventory methods are allowed under AS-2.
4. State the formula for WDV rate (from cost, salvage, life).
5. Differentiate capital expenditure, revenue expenditure, deferred revenue expenditure.

CHAPTER 6 — Non-Profit Organisations & Self-Balancing Ledgers

Importance: ★★★ MEDIUM (≈5 Qs / paper) **Difficulty:** Easy-Medium. Mechanical conversion + 2-3 facts about self-balancing ledgers.

6.0 — Understanding Non-Profit Accounts from First Principles

QUICK RECALL

How are NPO accounts different — and why?

A school, hospital trust, sports club, or temple trust exists to serve a purpose, not to earn profit for an owner. Its financial goal is to cover costs from subscriptions, donations, and grants. There is no "owner's profit" — but there is a surplus (income > expenditure) or deficit (expenditure > income for the period).

Because the concept of profit is absent, NPOs cannot have a "Trading Account" or "Profit & Loss Account" in the ordinary sense. Instead, they prepare:

1. **Receipts & Payments Account (R&P):** A real account summarising ALL cash received and ALL cash paid during the period — whether capital or revenue in nature, whether related to current year or previous/future years. It is essentially a summary of the Cash Book. Prepared on cash basis.
2. **Income & Expenditure Account (I&E):** The NPO equivalent of a Profit & Loss Account. Only revenue (recurring) items; prepared on **accrual basis** — outstanding and prepaid items are adjusted. Shows **Surplus** (if income > expenditure) or **Deficit** (if expenditure > income).
3. **Balance Sheet:** Same as for profit organisations, but "Capital" is replaced by "**Capital Fund**" (or "General Fund") — the accumulated surplus of past years.

Key conversion point: Receipts & Payments is cash-based (includes capital items). Income & Expenditure is accrual-based (excludes capital items, includes outstanding/prepaid). Converting between them is a common exam question.

Converting Subscription receipts (R&P) to Subscription income (I&E):

Income from Subscriptions = Subscriptions received (R&P) + Outstanding for current year – Advance received for next year – Outstanding from last year

Solved Example 6.A: A club received ₹60,000 as subscriptions in the year. Outstanding subs at year-end = ₹5,000. Advance subs received for next year = ₹3,000. Outstanding from last year still uncollected = ₹2,000 (received this year). Calculate subscription income for I&E.

*Subscriptions received: ₹60,000 Add: outstanding at year-end: +₹5,000 Less: advance for next year: –₹3,000 Less: last year's outstanding collected this year: –₹2,000 (this was last year's income) **Subscription income for I&E = ₹60,000***

6.1 Examiner mindset

Angle	Pot question
Receipts & Payments A/c features	"It is a summary of the cash book — T/F" → True
Income & Expenditure A/c	What goes in vs Receipts & Payments
Capital fund of NPO	= Surplus assets over liabilities (no profit motive)
Subscription accounting	Dues outstanding / advance handling
Self-Balancing Ledgers	Three ledgers maintained
Adjustment ledger / control account	Concepts

6.2 Non-Profit Organisations — terminology

For-profit	Non-profit
Capital	Capital Fund (or Accumulated Fund / General Fund)
Profit	Surplus
Loss	Deficit
Profit & Loss A/c	Income & Expenditure A/c
Cash book	Receipts & Payments A/c

6.3 Receipts & Payments vs Income & Expenditure

Aspect	Receipts & Payments A/c	Income & Expenditure A/c
Nature	Real account (cash summary)	Nominal account
Records	Only cash transactions of all years	Income/expense of current year only
Capital vs Revenue	Both included	Only revenue items
Period	Includes prior + current + advance	Strictly current year (accrual basis)
Opening / closing balance	Yes (cash + bank balances)	No — shows surplus / deficit only
Used by	Beginning of conversion	Final result statement

6.4 Conversion — Receipts & Payments → Income & Expenditure

Step	What to do
1	Take Receipts & Payments A/c
2	Remove capital items (e.g., legacy received, donations for a building)
3	Adjust each revenue item for outstanding & advance (accrual basis)
4	Add depreciation, write-offs, provisions
5	Compute net surplus/deficit

Subscription accounting (the favourite question)

Subscription **due** in current year (the figure for I&E) = Subscription received during year + **outstanding at year-end** – **outstanding at beginning** – **advance at year-end** + **advance at beginning**

(Always work it through; the formula is sign-heavy and examiners rotate the wording.)

6.5 Self-Balancing Ledgers

A system where the principal ledger is split into 3 self-contained sub-ledgers, each independently balanceable.

Ledger	Used for
General Ledger	Real and nominal accounts
Debtors (Sales) Ledger	Personal accounts of all customers
Creditors (Bought / Purchase) Ledger	Personal accounts of all suppliers

Each sub-ledger has an **Adjustment Account** (also called **Control Account**) which mirrors the totals to make each ledger balance independently.

Adjustment A/c	In which ledger	Purpose
Total Debtors A/c (or Sales Ledger Adj.)	General Ledger	Acts as control on debtors ledger
Total Creditors A/c (or Bought Ledger Adj.)	General Ledger	Acts as control on creditors ledger
General Ledger Adjustment A/c	Sales / Bought ledger	Mirrors above in the sub-ledgers

Advantages: localisation of errors, division of labour, helps preparing trial balance from each sub-ledger separately.

6.6 Worked PYQ-style examples

Q 6.1. A receipt of ₹50,000 as legacy. Where recorded?

Capital nature → **directly added to Capital Fund** (Balance Sheet), not in I&E.

Q 6.2. Subscription received ₹10,000; outstanding at year-start ₹500; outstanding at year-end ₹800. Subscription for I&E?

- = 10,000 + 800 (closing OS) – 500 (opening OS)
- = **₹10,300**

Q 6.3. Subscription received ₹15,000 (includes ₹1,000 received in advance for next year). Subscription for I&E?

15,000 – 1,000 = **₹14,000**.

Q 6.4. Receipts & Payments A/c is essentially a:

Summary of the cash book for the period.

Q 6.5. A building bought by a club for ₹2,00,000 — recorded in:

Capital nature → **Asset side of Balance Sheet**; not in I&E.

Q 6.6. Excess of Income over Expenditure is termed:

Surplus (added to Capital Fund).

Q 6.7. Self-balancing ledger system splits ledger into how many divisions?

Three (General, Debtors, Creditors).

Q 6.8. Total Debtors A/c is maintained in which ledger?

General Ledger.

Q 6.9. A sports club sells trophies received as donations. Treatment?

Sale = revenue receipt → I&E (Cr).

Q 6.10. Income & Expenditure A/c is prepared on:

Accrual basis.

6.7 Trap-recognition card

Trap	Defence
R&P A/c includes accruals	No — only cash actually received/paid
Capital donations in I&E	No — Balance Sheet (Capital Fund or specific fund)
Surplus = profit	Conceptually similar but term used in NPO is surplus
Depreciation in R&P A/c	No — non-cash, only in I&E
Subscription advance is income for current year	No — liability in B/S until year due

6.8 Mini-mock

#	Q	Answer
1	NPO uses ... instead of P&L?	Income & Expenditure A/c
2	NPO uses ... instead of capital?	Capital Fund
3	R&P A/c records?	Cash transactions of all years
4	Excess of expenditure over income = ?	Deficit
5	Self-balancing system has how many ledgers?	3
6	Adjustment A/c is also called?	Control A/c
7	Subscription due in current year formula?	Received + closing OS – opening OS – closing adv + opening adv
8	Donations for building → ?	Capital Fund (Asset side)
9	Excess of income over expenditure = ?	Surplus → added to Capital Fund
10	Sale of old newspapers in NPO is?	Revenue receipt (I&E Cr)

6.9 Active-recall prompts

1. State three differences between R&P A/c and I&E A/c.
2. Write the subscription-conversion formula.
3. Name three self-balancing sub-ledgers.
4. Give 3 advantages of self-balancing system.
5. State NPO terminology equivalents (capital, profit, loss, P&L).

=== PART B — ECONOMICS & GOVERNANCE (60 Qs / 120 marks) ===

CHAPTER 7 — CAG of India + Finance Commission

Importance: ★★★★★ CRITICAL (≈6 Qs / paper) **Difficulty:** Easy. Pure factual + constitutional. Score full marks here.

7.0 — Understanding CAG & Finance Commission from First Principles

QUICK RECALL

Who audits the government? Why must the auditor be independent?

In a private company, shareholders appoint an external auditor to verify that management hasn't misused funds. Citizens are the "shareholders" of a democracy — they need someone to audit the government's use of public money on their behalf. But citizens can't audit government accounts themselves. That agent is the **Comptroller and Auditor General of India (CAG)**.

The CAG's independence is constitutionally guaranteed precisely because the government cannot be allowed to control its own auditor — that would make the audit meaningless. Three protection mechanisms: (1) appointed by the President (not by Cabinet), (2) can only be removed by impeachment (same as a Supreme Court judge), (3) salary and service conditions charged to the Consolidated Fund of India (so Parliament cannot reduce the salary as a threat).

The Finance Commission solves a structural federal problem: India's Constitution gives the Centre the power to collect most taxes (income tax, corporate tax, GST). But States provide most public services (education, health, police, local infrastructure). Without a mechanism to share central taxes with States, the States would be permanently resource-starved. The Finance Commission (Article 280) is the constitutional referee — appointed every 5 years — that recommends how central taxes are divided vertically (Centre vs States collectively) and horizontally (among the 28 States).

CAG — key constitutional facts:

Fact	Detail
Constitutional Article	Article 148
Appointment	By the President of India
Removal	By Parliament (same process as Supreme Court judge — address by both Houses)
Term	6 years or until age 65 , whichever is earlier
Re-appointment	Not eligible for re-appointment
Salary charged to	Consolidated Fund of India (not voted by Parliament)
Audits	Union accounts, State accounts, Central PSUs, aided bodies
Reports submitted to	President (for Union) / Governor (for State) → laid before Parliament / Legislature
India's CAG is	Only an Auditor (not a Comptroller — unlike UK); cannot stop government spending

Finance Commission — key constitutional facts:

Fact	Detail
Constitutional Article	Article 280
Appointment	By the President of India
Frequency	Every 5 years (or earlier if President deems fit)
Composition	Chairman + 4 other members
Recommends	(i) Vertical devolution — % of Central taxes to States; (ii) Horizontal devolution — share among States; (iii) Grants-in-aid to States
15th FC	Chair: N.K. Singh; Period: 2021–26; States get 41% of divisible pool
16th FC	Chair: Arvind Panagariya; Period: 2026–31
Basis for horizontal distribution	Population, area, income distance, demographic performance, forest cover

7.1 Examiner mindset

Angle	Pet question
Article number for CAG	Art. 148
Tenure / oath / removal	"Tenure of CAG?" → 6 yrs or 65 yrs whichever earlier
Functions / duties of CAG	Audit of receipts and expenditure
Reports submitted to	President / Governor; placed before Parliament / State Legislature
Finance Commission article	Art. 280
FC composition	1 Chairman + 4 other members; appointed by President
Latest FC + period	15th FC (2021–26); 16th FC (2026–31)
FC functions	Tax devolution + grants-in-aid + measures to augment Consolidated Fund of States

7.2 Comptroller & Auditor General (CAG)

Item	Detail
Constitutional Article	Art. 148
Appointed by	President of India (under his hand and seal)
Oath	Administered by President
Tenure	6 years OR until age 65 , whichever is earlier
Removal	Same procedure as a Supreme Court judge (resolution of both Houses, 2/3 majority on grounds of proven misbehaviour or incapacity)
Salary	Charged on Consolidated Fund of India (not voted)
Conditions of service	Cannot hold any office under GoI or any State government after retirement
Reports submitted to	President (Union) → Parliament; Governor (State) → State Legislature
Reports examined by	Public Accounts Committee (PAC), Committee on Public Undertakings (COPU), Estimates Committee
Other names / role	"Guardian of the public purse" (Dr. B.R. Ambedkar)

Functions and duties (DPC Act, 1971)

1. Audits **all receipts and expenditure** from the Consolidated Fund of India and States.
2. Audits all expenditure from **Contingency Fund** and **Public Account**.
3. Audits **all trading, manufacturing, profit & loss accounts** of any department of GoI or State.
4. Audits accounts of **government companies** (under Companies Act).
5. Audits accounts of **statutory corporations** (when entrusted by law).
6. Audits accounts of any **autonomous body / authority** substantially financed by GoI/States.
7. Advises President on the form in which accounts shall be kept.
8. Submits **3 audit reports** annually to the President: (a) Audit Report on Appropriation Accounts, (b) Audit Report on Finance Accounts, (c) Audit Report on Public Undertakings.

Role: Comptroller vs Auditor

Comptroller (control)	Auditor (review)
Authorise withdrawal of money from Consolidated Fund	Audit expenditure already incurred
Indian CAG performs only the AUDIT function ; control over withdrawals lies with the executive (no comptroller function in India unlike UK)	This is a famous one-line trap

7.3 Finance Commission (FC)

Item	Detail
Constitutional Article	Art. 280
Constituted by	President of India every 5 years (or earlier if needed)
Composition	1 Chairman + 4 other members
Qualifications (Chairman)	Person with experience in public affairs
Qualifications (members)	Judge of HC, person with finance/audit/economics expertise
Tenure	Specified in order; usually period of recommendations (5 yrs)
Status	Quasi-judicial body
First FC	1951 (K.C. Neogy as Chairman)
Current (15th FC)	Chairman: N.K. Singh ; Period: 2021-22 to 2025-26
16th FC	Chairman: Dr. Arvind Panagariya ; Period: 2026-27 to 2030-31

Functions of FC (Art. 280)

1. **Distribution of net proceeds of taxes** between Centre and States (vertical devolution).
2. **Allocation between States** of their respective shares (horizontal devolution).
3. **Principles governing grants-in-aid** to States out of the Consolidated Fund of India.
4. **Measures to augment the Consolidated Fund of a State** to supplement the resources of Panchayats and Municipalities (added by 73rd & 74th Amendments).
5. **Any other matter** referred by the President in the interest of sound finance.

Tax devolution percentages (recent FCs)

FC	Devolution to States	Period
13th	32 %	2010 - 15
14th	42 % (jump)	2015 - 20
15th	41 % (J&K becoming UT reduced base)	2021 - 26

7.4 Worked PYQ-style examples

Q 7.1. CAG is appointed under which Article?

Article 148.

Q 7.2. The CAG holds office for:

6 years or until 65 years, whichever is earlier.

Q 7.3. "Guardian of the public purse" refers to:

CAG (phrase by Dr. Ambedkar).

Q 7.4. CAG's salary is:

Charged on the **Consolidated Fund of India** (not subject to Parliamentary voting).

Q 7.5. The Finance Commission is constituted under Article:

280.

Q 7.6. Composition of FC:

1 Chairman + 4 members.

Q 7.7. The 15th Finance Commission's chairman:

N.K. Singh.

Q 7.8. 14th FC recommended what % of central tax pool to States?

42 %.

Q 7.9. Who appoints CAG?

President of India.

Q 7.10. Who can remove CAG?

Removal by President only on the basis of a resolution passed by **both Houses of Parliament** with special majority (same procedure as Supreme Court judge).

Q 7.11. CAG audits accounts of:

Centre, States, Union Territories, government companies, statutory corporations, and substantially financed autonomous bodies.

Q 7.12. Who examines CAG reports in Parliament?

Public Accounts Committee (PAC) primarily.

Q 7.13. Which body advises on principles of grants-in-aid to States from the CFI?

Finance Commission (Art. 280).

Q 7.14. First Finance Commission was constituted in:

1951.

Q 7.15. 16th Finance Commission's Chairman:

Arvind Panagariya (recommendations period 2026-27 to 2030-31).

7.5 Trap-recognition card

Trap	Defence
CAG performs both Comptroller and Auditor functions	No — only Auditor in India.
FC has 5 + 5 members	No — 1 Chairman + 4 members (5 total).
14th FC devolution = 32 %	No — 42 % (the famous jump).
FC is statutory body	No — Constitutional (Art. 280).
CAG removed by President alone	No — needs both Houses' resolution.

7.6 Mini-mock

#	Q	Answer
1	Article for CAG	148
2	Tenure of CAG	6 yrs / 65 yrs
3	FC chairman of 14th	Y.V. Reddy

#	Q	Answer
4	FC chairman of 15th	N.K. Singh
5	Article for FC	280
6	Devolution % under 15th FC	41 %
7	CAG salary charged on	Consolidated Fund of India
8	Reports of CAG examined by	PAC
9	First FC chairman	K.C. Neogy
10	Type of body — FC	Constitutional, quasi-judicial

7.7 Active-recall prompts

1. State Article numbers for CAG and FC.
2. List 5 functions of CAG.
3. List 4 functions of FC.
4. Name latest 3 FC chairmen with their devolution %.
5. Differentiate Comptroller vs Auditor; state which India's CAG performs.

CHAPTER 8 — Basic Concepts of Economics & Introduction to Microeconomics

Importance: ★★★ MEDIUM (≈4 Qs / paper) **Difficulty:** Easy. Definitions + central problems.

8.0 — Understanding Basic Economics from First Principles

QUICK RECALL

What is economics? What fundamental problem does it study?

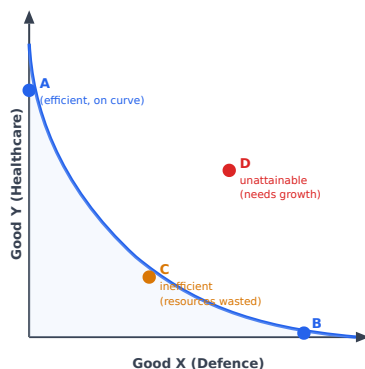
Economics is the study of how individuals, firms, and societies allocate **scarce resources** to satisfy **unlimited wants**. Resources — land, labour, capital, entrepreneurship — are finite. Human desires are not. This mismatch is the fundamental economic problem: **scarcity**. Every economic decision is a choice about what to do with limited resources, implying a **trade-off** and an **opportunity cost** (the value of the best alternative forgone).

Lionel Robbins' classic definition (1932): *"The science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."*

Three central economic questions every society must answer: 1. **What to produce?** Guns or butter? Schools or highways? (Allocation problem) 2. **How to produce?** Labour-intensive or capital-intensive? (Efficiency problem) 3. **For whom to produce?** Who gets how much of what is produced? (Distribution problem)

Micro vs Macro — two lenses on the same economy: - **Microeconomics:** Examines individual units — a household choosing between rice and wheat, a firm deciding its output, a market setting a price. "Micro" = zoom in. - **Macroeconomics:** Examines the economy as a whole — national income, inflation, unemployment, exchange rates. "Macro" = zoom out.

Production Possibility Frontier (PPF) — the opportunity cost diagram:



- **Movement along PPF (A→B):** More defence, less healthcare — opportunity cost of defence is healthcare forgone.

Point inside PPF: Resources wasted (unemployment, inefficiency).

Point outside PPF: Impossible with current resources — becomes possible only with economic growth (PPF shifts outward).

Outward shift of PPF: Caused by technological improvement, increase in resources, better education/skills.

Economic systems — three main types:

System	Decision-maker	Example	Characteristic
Capitalist / Market	Private individuals via price mechanism	USA, UK	Efficient but can be unequal
Socialist / Command	State central planning	Former USSR	Equality-focused but inefficient
Mixed (India's system)	State + market together	India , France	Balance of efficiency and welfare

Solved Example 8.A: "Opportunity cost of going to college full-time for a student who was earning ₹3 lakh/year" — what is it?

₹3 lakh per year in foregone salary (the income they gave up). Plus any tuition costs. **Opportunity cost = value of the best forgone alternative.**

8.1 Examiner mindset

Angle	Pet question
Definitions of economics	Adam Smith / Marshall / Robbins / Samuelson
Branches	Micro vs Macro
Central problems of an economy	What / How / For whom
Production Possibility Curve (PPC)	Concept and shape
Opportunity cost	Definition + example
Positive vs Normative economics	"Should be" vs "is"

8.2 Definitions of Economics

Economist	Definition (school)
Adam Smith	"Science of Wealth" (1776 — <i>Wealth of Nations</i>)
Alfred Marshall	"Study of mankind in the ordinary business of life" (Welfare definition)
Lionel Robbins	"Science which studies human behaviour as a relationship between ends and scarce means which have alternative uses" (Scarcity definition — most used)
Paul Samuelson	Modern definition — economics as the science of choice and resource allocation under scarcity, also examining growth

Robbins' four pillars: 1. Unlimited wants 2. Scarce means 3. Alternative uses of means 4. Need for choice

8.3 Branches of economics

Branch	Focus	Examples
Microeconomics	Individual units (consumer, firm, market)	Demand of one product, pricing in a single market
Macroeconomics	Aggregates (national income, employment, inflation)	GDP, unemployment, monetary policy
Coined by	Ragnar Frisch (1933)	

8.4 Central problems of an economy

Problem	Question	Concerns
What to produce	Which goods, in what quantity?	Allocation of resources
How to produce	Labour-intensive or capital-intensive technique?	Choice of technique
For whom to produce	Whose needs? Rich or poor?	Distribution of national income
(Modern additions)	Efficient use? Full employment? Growth?	Efficiency, employment, growth

8.5 Production Possibility Curve (PPC) / Frontier (PPF)

PPC = locus of all combinations of two goods that an economy can produce by fully and efficiently using its given resources and technology.

Properties

#	Property	Reason
1	Slopes downward	More of one good $\Rightarrow$ less of the other (scarcity)
2	Concave to the origin	Increasing opportunity cost (resources are not equally productive in both uses)
3	Points on the curve	Efficient and feasible
4	Points inside the curve	Feasible but inefficient (under-utilisation / unemployment)
5	Points outside the curve	Infeasible (need more resources or better technology)
6	Outward shift	Economic growth (more resources / better tech)

8.6 Opportunity cost

Opportunity cost = value of the next best alternative foregone when a choice is made.

Example: A student spends 1 hour on movie instead of study $\rightarrow$ opportunity cost = the marks lost from not studying for that hour.

In production: To produce 1 more unit of guns on the PPC, the economy must sacrifice some units of butter — that sacrifice is the opportunity cost.

8.7 Positive vs Normative Economics

Positive	Normative
Describes "what is"	Prescribes "what ought to be"
Objective, fact-based	Value-based, subjective
"Inflation in India is 6 %"	"Inflation should be reduced to 4 %"
Testable	Not testable

8.8 Worked PYQ-style examples

Q 8.1. Robbins defined economics as the study of:

Behaviour relating to **ends** and **scarce means** with alternative uses.

Q 8.2. "Wealth of Nations" was written by:

Adam Smith (1776).

Q 8.3. Microeconomics deals with:

Individual units like consumer, firm, market.

Q 8.4. Macro & Micro terms coined by:

Ragnar Frisch (1933).

Q 8.5. Central problem "How to produce" relates to:

Choice of **technique** (labour vs capital intensive).

Q 8.6. PPC is concave to origin because of:

Increasing opportunity cost (Marginal Rate of Transformation rises).

Q 8.7. A point inside the PPC indicates:

Underemployment of resources / inefficiency.

Q 8.8. Normative economics deals with:

"What ought to be" — value judgements.

Q 8.9. "Father of Economics":

Adam Smith.

Q 8.10. Fundamental cause of all economic problems:

Scarcity of resources relative to wants.

8.9 Trap-recognition card

Trap	Defence
Marshall = scarcity definition	No, Marshall = welfare; Robbins = scarcity.
PPC is convex	No — concave to origin (increasing opp cost).
Macroeconomics = study of inflation only	No — also output, employment, BoP, growth.
All economics is normative	No — positive economics is fact-based.

8.10 Mini-mock

#	Q	Answer
1	Father of economics?	Adam Smith
2	Scarcity definition?	Robbins
3	Welfare definition?	Marshall
4	Coined micro/macro?	Ragnar Frisch
5	Central problems?	What, How, For whom
6	PPC shape?	Concave to origin
7	Outside PPC = ?	Infeasible
8	Inside PPC = ?	Underutilisation
9	Outward shift of PPC?	Economic growth
10	Normative example?	"Income should be equal"

8.11 Active-recall prompts

1. State Robbins' definition.
2. List 5 properties of PPC.
3. Define opportunity cost with one example.
4. Differentiate positive vs normative.
5. Name 3 central problems of an economy.

CHAPTER 9 — Theory of Demand & Supply (incl. Elasticity, Consumer Behaviour)

Importance: ★★★★★ CRITICAL (≈8 Qs / paper) **Difficulty:** Medium. Most-tested chapter in Part B.

9.0 — Understanding Demand, Supply & Elasticity from First Principles

QUICK RECALL

Why does the demand curve slope downward? It's two effects working together.

When the price of a good rises, people buy less of it — this is the **Law of Demand**. But why? Two reasons work simultaneously:

1. **Substitution effect:** When the price of onions rises, onions become relatively more expensive compared to other vegetables. Consumers switch to tomatoes, capsicum, etc. — they substitute away from the now-costlier good.
2. **Income effect:** When the price of onions rises, the consumer's real purchasing power falls (the same income buys less). They cut back on consumption across the board, including onions.

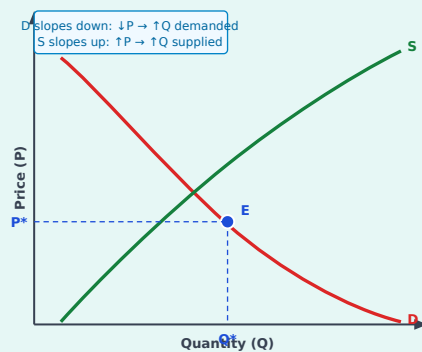
Both effects push demand down when price rises — hence the downward-sloping demand curve.

Demand curve shifts vs movements along the curve — the most-tested distinction:

- A **change in price** causes movement *along* the existing demand curve (extension if price falls, contraction if price rises).
- A **change in anything else** (income, tastes, price of related goods, expectations) causes the entire curve to *shift* left or right.

Why does the supply curve slope upward? A higher price makes production more profitable → existing producers expand output → new producers enter → total quantity supplied rises. Hence the direct (positive) relationship between price and quantity supplied.

Equilibrium is where supply meets demand. At the equilibrium price: quantity demanded = quantity supplied. No surplus (excess supply), no shortage (excess demand). Price adjusts automatically: surplus → price falls → demand rises, supply falls, until equilibrium restored; shortage → price rises → demand falls, supply rises, until restored.



Elasticity — the measure of responsiveness:

FORMULA

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

(Always negative for normal goods — we conventionally report $|E_d|$)

The five elasticity values and their revenue implications:

$ E_d $	Type	Example	When price rises, total revenue...
> 1	Elastic	Luxury goods, many substitutes	Falls (demand falls more than price rises)
< 1	Inelastic	Necessities, few substitutes (salt, insulin)	Rises (demand falls less than price rises)
$= 1$	Unit elastic	—	Unchanged
$= 0$	Perfectly inelastic	Life-saving drug with no substitute	Rises proportionally (demand unchanged)
$= \infty$	Perfectly elastic	Perfect competition commodity	Falls to zero (any price rise → all demand lost)

Solved Example 9.A — Calculating price elasticity:

Price rises from ₹10 to ₹12 (20% increase); quantity demanded falls from 100 to 80 (20% decrease).

$$E_d = \frac{-20\%}{+20\%} = -1.0 \text{ (unit elastic, } |E_d| = 1)$$

Solved Example 9.B — Revenue implication:

A monopolist faces inelastic demand ($|E_d| = 0.5$). If it raises price by 10%, what happens to total revenue?

*Inelastic demand: quantity falls by only $0.5 \times 10\% = 5\%$. Revenue change = Price effect (+10%) – Quantity effect (–5%) → **Revenue rises by approximately 5%.***

Consumer surplus — the "benefit beyond the price paid":

Consumer surplus = the area below the demand curve and above the market price line. It represents the total value consumers get beyond what they pay. A fall in price increases consumer surplus. Monopoly reduces consumer surplus (charges higher price, produces less). This loss is the "deadweight loss of monopoly."

9.1 Examiner mindset

Angle	Pet question
Law of demand and its exceptions	Giffen / Veblen / Speculative goods
Determinants of demand and supply	Memorise lists
Elasticity types	Price (own/cross), Income
Numerical: $E_d = \% \Delta Q / \% \Delta P$	Direct compute
Consumer behaviour: Marshallian (Cardinal) vs Indifference (Ordinal)	Difference in approach
Equilibrium price	At intersection of D and S

9.2 Law of Demand

Law of demand: *Other things being equal, as the price of a good rises, the quantity demanded falls (and vice versa). Price and quantity demanded are inversely related.*

Determinants of demand

1. **Price** of the good (own price)
2. **Income** of consumer
3. **Prices of related goods** (substitutes & complements)
4. **Tastes & preferences**
5. **Population / number of consumers**
6. **Expectations** (of future price/income)
7. **Government policy** (taxes, subsidies)

Exceptions to law of demand

Exception	What happens
Giffen goods (inferior food staples)	Price ↑ → demand ↑ (because real income falls and the consumer substitutes more of the cheaper inferior good)
Veblen goods (luxury / status)	Price ↑ → demand ↑ (snob/status appeal)
Speculative goods (stocks, real estate)	Price ↑ → demand ↑ (expect further rise)
Necessities (salt, food in famine)	Demand stays high regardless of price
Habits / addictions	Inelastic; doesn't follow the law

9.3 Movement vs Shift of demand curve

Change	What happens to curve
Change in price of the good	Movement along the curve (upward = contraction; downward = extension)
Change in non-price determinants (income, tastes, etc.)	Shift of the curve (rightward = increase; leftward = decrease)

9.4 Elasticity of Demand

(i) Price Elasticity of Demand (E_d)

$$E_d = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q / Q}{\Delta P / P}$$

By convention E_d is reported as **positive** (the inverse relationship is taken as given).

| $|E_d|$ | Type | Slope of curve | |-----|-----|-----| | 0 | Perfectly inelastic | Vertical | | < 1 | Inelastic | Steep | | = 1 | Unitary elastic | Rectangular hyperbola | | > 1 | Elastic | Flat | | ∞ | Perfectly elastic | Horizontal |

Methods of measurement

1. **Percentage method** (above)
2. **Total expenditure method:** If $P \uparrow$ and $TE \downarrow \rightarrow$ elastic; If TE constant $\rightarrow$ unitary; If $TE \uparrow \rightarrow$ inelastic
3. **Geometric (point) method:** At a point on a straight-line demand curve, E_d = lower segment / upper segment

4. **Arc method:** Average of two prices and quantities for a discrete change

(ii) Income Elasticity of Demand (E_y)

$$E_y = \frac{\% \Delta Q}{\% \Delta Y}$$

E_y	Good type
> 0	Normal good
> 1	Luxury (income elastic)
$0 < E_y < 1$	Necessity (income inelastic)
< 0	Inferior good
= 0	Neutral (insensitive to income)

(iii) Cross Elasticity of Demand (E_{xy})

$$E_{xy} = \frac{\% \Delta Q_x}{\% \Delta P_y}$$

E_{xy}	Goods relationship
> 0	Substitutes (e.g., tea & coffee — P_y up → Q_x up)
< 0	Complements (e.g., car & petrol — P_y up → Q_x down)
= 0	Unrelated

9.5 Consumer Behaviour — two approaches

Marshallian (Cardinal Utility) approach

Utility is **measurable** in numerical units (utils). Consumer maximises total utility.

Law of Diminishing Marginal Utility (DMU): As consumption of a good increases, the marginal utility derived from each additional unit decreases.

Equilibrium condition: $\frac{MU_x}{P_x} = \frac{MU_y}{P_y} = \dots = \frac{MU_m}{1}$ (Law of Equi-marginal Utility)

Indifference Curve (Ordinal Utility) approach

Utility cannot be measured but bundles can be **ranked** (ordinal). An indifference curve shows all combinations of two goods giving the consumer **equal satisfaction**.

Properties of IC:

#	Property
1	Slopes downward (more of one ⇒ less of other for same satisfaction)
2	Convex to origin (Diminishing Marginal Rate of Substitution — DMRS)
3	Higher IC → higher satisfaction
4	Two ICs never intersect
5	Doesn't touch axes

Marginal Rate of Substitution (MRS):

$$MRS_{xy} = \frac{\Delta Y}{\Delta X} = \frac{MU_x}{MU_y}$$

Budget line:

$$P_x \cdot X + P_y \cdot Y = M$$

Equilibrium of consumer: Highest IC tangent to budget line. **Tangency condition:** $MRS_{xy} = P_x/P_y$ (slope of IC = slope of budget line).

9.6 Law of Supply

Law of supply: Other things being equal, as price rises, quantity supplied rises (and vice versa). Price and quantity supplied are **directly** related.

Determinants of supply

1. Price of the good
2. Prices of factors of production (input cost)

3. Technology
4. Number of producers
5. Government policy (taxes, subsidies)
6. Goal of the producer
7. Expectations of future price

Elasticity of supply

$$E_s = \frac{\% \Delta Q_s}{\% \Delta P}$$

Same numerical types as demand: 0 (perfectly inelastic, vertical), < 1, = 1, > 1, ∞ (perfectly elastic, horizontal).

9.7 Market Equilibrium

*Equilibrium = the price at which **quantity demanded = quantity supplied** (Qd = Qs). Graphically, the intersection of D and S curves.*

Effect	Equilibrium price	Equilibrium quantity
D shifts right (S unchanged)	↑	↑
D shifts left (S unchanged)	↓	↓
S shifts right (D unchanged)	↓	↑
S shifts left (D unchanged)	↑	↓
Both shift right	uncertain (depends on magnitude)	↑
D right + S left	↑	uncertain

9.8 Worked PYQ-style examples

Q 9.1. A 10 % increase in price of a good causes 20 % fall in demand. E_d ?

$|20/10| = 2$ (elastic).

Q 9.2. Cross elasticity between tea and sugar is:

Negative (complements).

Q 9.3. A normal good has income elasticity:

0 (positive).

Q 9.4. Giffen goods are an exception to:

Law of demand.

Q 9.5. Demand curve shifts due to:

Change in **non-price** determinants (income, tastes, prices of related goods, expectations, etc.).

Q 9.6. Indifference curves are:

Convex to origin (due to diminishing MRS).

Q 9.7. Equilibrium of consumer (cardinal):

$MU_x/P_x = MU_y/P_y = \dots$ (Equi-marginal utility).

Q 9.8. Equilibrium of consumer (ordinal):

$MRS_{xy} = P_x/P_y$ (highest IC tangent to budget line).

Q 9.9. A perfectly inelastic supply curve is:

Vertical.

Q 9.10. When demand curve shifts rightward and supply unchanged, equilibrium price:

Rises (and quantity rises).

Q 9.11. Total expenditure method: if $E_d = 1$, then on a price change, total expenditure:

Remains constant.

Q 9.12. Demand for salt is:

Highly **inelastic** ($E_d \approx 0$).

Q 9.13. If $E_y > 1$, the good is:

Luxury (income elastic).

Q 9.14. A 5 % rise in income raises demand by 8 %. Income elasticity?

$8/5 = 1.6$ (luxury).

Q 9.15. Two goods with $E_{xy} = +0.7$ are:

Substitutes (positive cross elasticity).

9.9 Trap-recognition card

Trap	Defence
Veblen goods break law of demand	True (price up = status up = demand up)
Inferior good = Giffen good	All Giffen are inferior, but not all inferior are Giffen
Movement along vs shift	Movement = price change; shift = non-price change
Negative cross-elasticity = substitutes	No — negative = complements
Demand curve always negatively sloped	Exceptions: Veblen, Giffen, speculative
Marshallian = ordinal	No — Marshallian = cardinal; IC = ordinal

9.10 Mini-mock

#	Q	Answer
1	Law of demand: $P \uparrow \rightarrow Q?$	Falls
2	$E_d = 0$ means?	Perfectly inelastic
3	$E_d = \infty$ means?	Perfectly elastic
4	Cross elasticity of complements?	Negative
5	Income elasticity of inferior good?	Negative
6	IC convex to origin because?	Diminishing MRS

#	Q	Answer
7	Two ICs intersecting violates which property?	Higher IC = higher utility (transitivity)
8	Equilibrium of consumer (ordinal)?	$MRS_{xy} = P_x/P_y$
9	Slope of budget line?	$-P_x/P_y$
10	Veblen good is exception to?	Law of demand

9.11 Active-recall prompts

1. State the law of demand and 4 exceptions.
2. List 7 determinants of demand.
3. Differentiate movement vs shift of demand curve.
4. Write the formula and types of price/income/cross elasticity.
5. State 5 properties of IC.

CHAPTER 10 — Theory of Production & Cost

Importance: ★★★★★ HIGH (≈6 Qs / paper) **Difficulty:** Medium. Memorise the 3 laws and the 7 cost concepts.

10.0 — Understanding Production & Cost Theory from First Principles

QUICK RECALL

How do firms decide how much to produce? What does "diminishing returns" really mean?

A firm uses inputs (labour, capital, raw materials) to produce outputs. The **production function** $Q = f(L, K)$ describes the maximum output achievable from given input combinations. It is a purely technical relationship — what is physically possible.

Short run vs long run: - **Short run:** At least one input is **fixed** (usually capital — factory buildings, machinery take time to expand or contract). Only variable inputs (typically labour) can be changed. - **Long run:** ALL inputs are variable. The firm can build new factories, buy new machines, enter or exit the industry.

Law of Diminishing Returns (short run): Add workers to a fixed factory. Initially, each extra worker adds a lot — they can specialise, divide tasks. But eventually the factory becomes crowded — extra workers get in each other's way, share tools, wait for machines. Each additional worker adds *less* extra output than the previous one. This is **diminishing marginal product**. This is a **short-run law** — it occurs only because capital is fixed.

Cost curves — why they are U-shaped: Average Cost (AC) falls initially because fixed costs are spread over more units as output rises. AC eventually rises because of diminishing returns — each extra unit requires more and more variable input. The U-shape is the combination of these two forces. **The MC curve always cuts the AC curve at its minimum** — this is a mathematical identity, not a coincidence (when marginal cost < average cost, it pulls the average down; when MC > AC, it pulls the average up; so they must cross at the minimum).

The seven cost concepts — relationships that examiners test:

FORMULA

$$TC = FC + VC \quad ; \quad AC = \frac{TC}{Q} = AFC + AVC \quad ; \quad MC = \frac{\Delta TC}{\Delta Q}$$

Key relationships: - $AFC = FC/Q$ — always falls as Q rises (fixed cost spread over more units — "spreading overhead") - AVC — U-shaped (first falls due to increasing returns, then rises due to diminishing returns) - AC — U-shaped, lies above AVC; the vertical gap = AFC (which narrows as Q rises) - **MC cuts both AVC and AC at their respective minimum points**

Returns to scale (long run):

Type	What happens	Example
Increasing returns to scale	Double inputs → more than double output	Large factories with economies of scale
Constant returns to scale	Double inputs → exactly double output	—
Decreasing returns to scale	Double inputs → less than double output	Management becomes too complex

Solved Example 10.A:

FC = ₹10,000. VC at Q=100 is ₹5,000. Find TC, AC, AFC, AVC at Q=100. Also, VC at Q=101 is ₹5,040. Find MC.

$TC = 10,000 + 5,000 = ₹15,000$ - $AC = 15,000/100 = ₹150$ - $AFC = 10,000/100 = ₹100$ - $AVC = 5,000/100 = ₹50$ - $TC(101) = 10,000 + 5,040 = ₹15,040$ - $MC = 15,040 - 15,000 = ₹40$ Note: MC (₹40) < AVC (₹50) → AVC is still falling at Q=100.

10.1 Examiner mindset

Angle	Pet question
Factors of production	4 classical; 5 modern
Production function	TP, AP, MP relationship
Law of variable proportions	3 stages; SRMP curve shape
Returns to scale	Long-run; 3 cases
Short-run vs long-run costs	Definitions
Cost concepts	TC, TFC, TVC, AC, AFC, AVC, MC
Relationship between AC and MC	MC cuts AC at AC's minimum
Economies of scale	Internal (5 types) and external

10.2 Factors of production

Factor	Reward	Examples
Land	Rent	Natural resources, mines, agricultural land
Labour	Wages / Salary	Human effort, mental + physical

Factor	Reward	Examples
Capital	Interest	Machinery, tools, plant, money capital
Entrepreneur	Profit	Risk-taker, organiser
(Modern addition: Knowledge / Technology)	Royalty / fees	Innovation, IP

10.3 Production function

A **production function** shows the relationship between physical inputs and physical outputs of a firm: $Q = f(L, K, N, E)$

Total, Average, Marginal Product

Concept	Formula
Total Product (TP)	Total output produced
Average Product (AP)	TP / L (output per unit of variable factor)
Marginal Product (MP)	$\Delta TP / \Delta L$ (change in TP from one more unit of input)

Relationship (very high-yield)

When MP	... AP is	TP behaviour
MP > AP	AP is rising	TP rising
MP = AP	AP is at its maximum	TP still rising
MP < AP	AP is falling	TP rising at decreasing rate
MP = 0	AP positive	TP at maximum
MP < 0	AP falling	TP falling

10.4 Law of Variable Proportions (Short-run, one variable input)

As successive units of a variable input are added to a fixed input, **TP first rises at increasing rate, then at diminishing rate, and finally falls.**

Three stages

Stage	What happens	MP behaviour	Producer's choice
I — Increasing returns	TP rises rapidly; AP rises	MP rises, reaches max	Don't stop here (AP still rising)
II — Diminishing returns	TP rises at decreasing rate	MP falls but positive; MP < AP	Operate here (rational stage)
III — Negative returns	TP falls	MP becomes negative	Avoid — adding input reduces output

The producer always operates in **Stage II** because:

Stage I → variable factor is underused (could add more)

Stage III → variable factor is overused (causes loss)

Stage II → optimal balance

10.5 Returns to Scale (Long-run, all inputs vary)

When all inputs are increased by a proportion, output may increase by:

Scenario	Name
More than proportionate	Increasing Returns to Scale (IRS)
Proportionate	Constant Returns to Scale (CRS)
Less than proportionate	Decreasing Returns to Scale (DRS)

Causes of IRS: economies of scale (specialisation, large-scale buying, technical economies). Causes of DRS: managerial limits, coordination problems.

10.6 Cost Concepts

Short-run cost classification

Cost	Formula	Behaviour
Total Fixed Cost (TFC)	Cost that doesn't vary with output (rent, salary of permanent staff, depreciation on plant)	Constant
Total Variable Cost (TVC)	Varies with output (raw material, wages, fuel)	Rises with Q

Cost	Formula	Behaviour
Total Cost (TC)	$TFC + TVC$	$TFC + TVC$
Average Fixed Cost (AFC)	TFC / Q	Falls continuously (curve is rectangular hyperbola)
Average Variable Cost (AVC)	TVC / Q	U-shaped
Average Cost (AC)	$TC / Q = AFC + AVC$	U-shaped
Marginal Cost (MC)	$\Delta TC / \Delta Q$	U-shaped, cuts AC and AVC at their minimum

Key relationships

1. **MC cuts AC at AC's minimum** — when $MC < AC$, AC is falling; when $MC > AC$, AC is rising; at the bottom, $MC = AC$.
2. **MC cuts AVC at AVC's minimum** (same logic).
3. **AFC** never reaches zero (asymptotic).
4. As $Q \uparrow$, **AFC** falls and pulls AC down initially; eventually AVC's rise overcomes it → AC turns up.
5. **TFC** appears as horizontal straight line; **AFC** as rectangular hyperbola.

Long-run cost

All costs are variable in long run. The **LAC (Long-run Average Cost)** is the **envelope** of all SAC curves. Often U-shaped due to economies and diseconomies of scale.

10.7 Economies of Scale

Type	Description
Internal economies	Arise within the firm
1. Technical	Larger plants more efficient
2. Managerial	Specialisation in management
3. Marketing	Bulk purchase, advertising spread
4. Financial	Cheaper finance for big firms
5. Risk-bearing	Diversification
External economies	Arise outside (industry/region growth: better infrastructure, skilled labour pool)

Diseconomies of scale (the limit)

- Managerial inefficiency
- Coordination costs
- Communication breakdown
- Industrial unrest

10.8 Worked PYQ-style examples

Q 10.1. Reward of land?

Rent.

Q 10.2. Reward of entrepreneurship?

Profit.

Q 10.3. Total product is maximum when:

MP = 0.

Q 10.4. A producer always operates in which stage?

Stage II of variable proportions (diminishing positive returns).

Q 10.5. Long-run returns to scale: 10 % input rise → 15 % output rise. Type?

Increasing Returns to Scale.

Q 10.6. AFC curve shape?

Rectangular hyperbola (continuously falling).

Q 10.7. MC cuts AC at:

AC's **minimum** point.

Q 10.8. TFC = 100, TVC = 200 at Q = 50. AC = ?

- $TC = 100 + 200 = 300$
- $AC = 300/50 = 6$

Q 10.9. When MP > AP:

AP is rising.

Q 10.10. Internal economies of large-scale production are also called:

Real economies (or simply internal economies).

Q 10.11. Specialisation of labour leads to:

Increasing returns / internal economies (managerial / technical).

Q 10.12. TC at Q = 10 is ₹500; at Q = 11 is ₹540. Marginal cost?

$540 - 500 = ₹40$.

Q 10.13. Long-run AC is the envelope of:

All short-run AC curves.

Q 10.14. When all inputs double and output doubles, returns to scale are:

Constant.

Q 10.15. Diseconomies of scale typically arise from:

Managerial inefficiency / coordination problems.

10.9 Trap-recognition card

Trap	Defence
Producer operates in Stage I	No — Stage II (rational)
AFC = 0 at high output	No — asymptotic
MC cuts AC at AC max	No — at AC minimum
Returns to scale and law of variable proportions are same	No — RtS is long-run (all inputs vary); LVP is short-run (one variable, one fixed)
Land's reward is wages	No — rent

10.10 Mini-mock

#	Q	Answer
1	TFC at Q=0 vs Q=100?	Same
2	TP max when MP = ?	0
3	AP max when MP = ?	AP
4	Long-run is when?	All inputs are variable
5	LRC envelope concept?	LAC envelopes SACs
6	MC at Q=20 if TC moves from ₹400 to ₹420?	₹20
7	4 factors of production?	Land, Labour, Capital, Entrepreneur

#	Q	Answer
8	Returns to scale measured in?	Long run
9	Specialisation gives which economy?	Internal (managerial / technical)
10	Cost that doesn't vary with output?	Fixed cost (TFC)

10.11 Active-recall prompts

1. State the law of variable proportions and its 3 stages.
2. List 7 cost concepts.
3. State the AC-MC relationship (3 cases).
4. Differentiate returns to scale vs returns to a factor.
5. List 5 internal economies of scale.

CHAPTER 11 — Forms of Market & Price Determination

Importance: ★★★★★ HIGH (≈6 Qs / paper) **Difficulty:** Medium. Master the four market structures and you score full.

11.0 — Understanding Market Structures from First Principles

QUICK RECALL

Why does a monopolist charge more than a competitive firm? And how much can it charge?

In **perfect competition**, there are many firms selling an identical product. No single firm can influence the price — if it charges even ₹1 more, all customers switch to competitors. The firm is a **price taker**. Its demand curve is perfectly horizontal (infinitely elastic). In the long run, profit = 0 (new entrants compete it away).

In a **monopoly**, there is only one seller. The firm IS the market. To sell more units, it must lower the price (because it faces the downward-sloping market demand curve). This gives the monopolist **price-setting power (pricing power)**. But it is not unlimited power — it can only charge what the market demand curve allows.

A monopolist maximises profit by producing where $MR = MC$ (same rule as any firm). But for a monopolist, the Marginal Revenue (MR) is always **below** the price — because to sell one more unit, the monopolist must lower the price on ALL units, not just the marginal one. So $P > MR$ for a monopolist. This means the monopolist produces LESS and charges MORE than a competitive market would → **deadweight loss** (society loses the triangle of welfare that neither producer nor consumer captures).

Oligopoly (few firms) is the most common real-world structure (telecom, aviation, FMCG, banking). Its defining feature: **strategic interdependence** — each firm's output and pricing decision directly affects and is affected by rivals' decisions. This makes oligopoly analysis complex and creates incentives for both competition and collusion.

Comparison of four market structures:

Feature	Perfect Competition	Monopoly	Oligopoly	Monopolistic Competition
Number of sellers	Very many	One	Few (2–10)	Many
Product	Identical (homogeneous)	Unique, no substitute	Identical or differentiated	Differentiated
Entry/exit	Perfectly free	Completely blocked	Restricted	Relatively free
Price control	None (price taker)	Full	Partial (interdependent)	Some (product differentiation)
Long-run profit	Zero ($P = AC$)	Can be positive	Can be positive	Zero ($P = AC$)
MR vs P	$MR = P$ (flat demand curve)	$MR < P$	$MR < P$	$MR < P$
Examples	Agricultural markets (wheat, rice)	Railways, BSNL (historically)	Aviation, telecom	Restaurants, toothpaste brands

Solved Example 11.A: A monopolist has demand $P = 100 - Q$ and $MC = 20$. Find profit-maximising output, price, and the deadweight loss region.

$MR = 100 - 2Q$ (for linear demand, MR has double the slope). Set $MR = MC$: $100 - 2Q = 20 \rightarrow Q^* = 40$. Price: $P^* = 100 - 40 = ₹60$. Competitive outcome: Set $P = MC$: $100 - Q = 20$ - Solving: $Q_c = 80$ - $P_c = 100 - 80 = ₹20$ Monopoly produces 40 (not 80) and charges ₹60 (not ₹20) — classic monopoly outcome.

11.1 Examiner mindset

Angle	Pet question
Features of perfect competition	Many buyers/sellers, homogeneous, free entry/exit, perfect knowledge
Monopoly features	Single seller, no close substitute, price-maker
Monopolistic competition	Many sellers, product differentiation
Oligopoly	Few sellers, mutual interdependence
Equilibrium of firm	$MR = MC$ condition
Price discrimination	Conditions for it to work
Kinked demand curve (oligopoly)	Sweezy's model

11.2 The 4 forms of market — comparison table

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Number of sellers	Very many	Many	Few	One
Number of buyers	Very many	Many	Many	Many
Product type	Homogeneous	Differentiated (close substitutes)	Homogeneous or differentiated	Unique (no close substitute)
Entry / exit	Free	Relatively free	Restricted (entry barriers)	Blocked

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
Knowledge	Perfect	Imperfect	Imperfect	Imperfect
Price control	None (price-taker)	Some	Considerable	Full (price-maker)
Demand curve	Perfectly elastic (horizontal)	Highly elastic (downward sloping, gentle)	Kinked	Inelastic, downward sloping
AR = MR?	Yes (AR = MR = price)	No (AR > MR)	No	No (AR > MR)
Equilibrium condition	MR = MC = P	MR = MC	MR = MC	MR = MC
Long-run profit	Normal only	Normal only	Super-normal (often)	Super-normal
Examples	Agriculture (closest), stock market	Restaurants, soaps, toothpastes	Auto, telecom, airlines	Indian Railways (historically), local water utility

11.3 Perfect Competition

Features (memorise the 6)

1. **Large number of buyers & sellers** — no individual can affect price
2. **Homogeneous product** — perfect substitutes
3. **Free entry & exit**
4. **Perfect knowledge** of market
5. **Perfect mobility** of factors
6. **No transport costs** (assumption)

Equilibrium

*Firm equilibrium: **MR = MC** (and MC rising). Industry equilibrium: where market demand = market supply.*

In **short run**: firm can earn super-normal profit, normal profit, or even loss. In **long run**: only **normal profit** (because free entry/exit erodes super-normal and forces out loss-makers).

11.4 Monopoly

Features

1. **Single seller**, many buyers
2. **No close substitute** for the product
3. **Strong barriers to entry** (legal, technological, control over input)
4. **Price-maker** (chooses price; faces downward-sloping demand)
5. **Average revenue (AR) > Marginal revenue (MR)** — both downward sloping; MR lies below AR

Equilibrium

MR = MC, with MC cutting MR from below. Monopolist sets price from the AR curve at the corresponding quantity.

Price discrimination

*Charging **different prices** to different buyers for the same good.*

Type	Description	Example
First-degree (perfect)	Different price to each unit	Auctions
Second-degree	Block pricing (per slab)	Electricity tariff
Third-degree	Different prices to different markets	Doctor's fee — rich vs poor

Conditions necessary: 1. Seller has market power (monopoly). 2. Markets must be **separable** (no resale arbitrage). 3. Demand elasticity must differ between markets.

11.5 Monopolistic Competition

Features

1. **Large number** of sellers
2. **Product differentiation** (real or imaginary — branding, packaging, quality)
3. Free entry / exit in long run
4. Selling cost / advertisement matters

5. **Non-price competition** is significant

Edward Chamberlin developed this theory (1933).

Equilibrium

- Short run: super-normal profit possible.
- Long run: only normal profit (entry erodes super-normal).
- $AR > MR$; both slope downward.

Excess capacity is a hallmark — firms produce **less than the optimum** (LAC minimum point), leading to higher AC than under perfect competition.

11.6 Oligopoly

Features

1. **Few sellers** (2 = duopoly, 3-10 = oligopoly typically)
2. **Mutual interdependence** — each firm's decision affects others' decisions
3. **Barriers to entry** (often)
4. **Price rigidity** (sticky prices)
5. **Heavy advertising / non-price competition**
6. Products can be homogeneous (cement, steel) or differentiated (cars)

Kinked Demand Curve (Sweezy's model)

*Explains **price rigidity in oligopoly**.*

*Each firm assumes: - If it **raises** price, **other firms will not follow** → demand falls drastically (elastic upper portion). - If it **lowers** price, **other firms will follow** → demand rises only slightly (inelastic lower portion).*

The kink at the prevailing price → MR curve has a **discontinuity** → MC can change within this gap without altering price → prices are sticky.

Other oligopoly models

- **Cournot** (quantity competition)

Bertrand (price competition)

Stackelberg (leader-follower)

Cartel (joint profit max — like OPEC)

Price leadership (dominant firm sets price; others follow)

11.7 Worked PYQ-style examples

Q 11.1. In which market form is $AR = MR$ throughout?

Perfect competition (because price is constant).

Q 11.2. A monopolist's MR curve:

Lies **below** the AR curve and slopes downward more steeply.

Q 11.3. Long-run equilibrium of a perfectly competitive firm earns:

Normal profit only.

Q 11.4. Product differentiation is the main feature of:

Monopolistic competition.

Q 11.5. Mutual interdependence is characteristic of:

Oligopoly.

Q 11.6. The kinked demand curve was developed by:

Paul Sweezy (1939).

Q 11.7. Equilibrium condition of a firm under any market is:

$MR = MC$ (with MC rising).

Q 11.8. Indian Railways is closest to which market form?

Monopoly (historically; partial liberalisation has changed this).

Q 11.9. A firm's demand curve in perfect competition is:

Horizontal (perfectly elastic) at the market price.

Q 11.10. Excess capacity is a feature of:

Monopolistic competition.

Q 11.11. Price discrimination requires:

Market power + separable markets + different elasticities.

Q 11.12. Selling cost / advertisement is most prominent in:

Monopolistic competition and oligopoly.

Q 11.13. Which market form has the most elastic demand curve?

Perfect competition (perfectly elastic = horizontal).

Q 11.14. Total revenue (TR) of a firm = ?

$Price \times Quantity = AR \times Q = \int MR \, dQ$.

Q 11.15. A cartel is associated with:

Oligopoly (joint profit maximisation by colluding firms).

11.8 Trap-recognition card

Trap	Defence
Monopoly always earns super-normal profit	Possible in long-run, not guaranteed in short-run (could earn normal).
Monopolistic competition = monopoly	No — many sellers in monopolistic competition.
Cartel is found in perfect competition	No — oligopoly.
Equilibrium condition only for monopoly	$MR = MC$ applies to all market forms.
Kinked demand curve = monopolistic competition	No — oligopoly.

11.9 Mini-mock

#	Q	Answer
1	Number of sellers in perfect competition?	Very large
2	Number in monopoly?	One
3	Demand curve slope in PC?	Horizontal
4	Demand curve in monopoly?	Downward sloping
5	AR vs MR in monopoly?	$AR > MR$
6	Long-run profit in monopolistic competition?	Normal only
7	Kinked demand by?	Sweezy
8	Price-maker means?	Can set price; monopoly

#	Q	Answer
9	Mutual interdependence is characteristic of?	Oligopoly
10	Excess capacity feature of?	Monopolistic competition

11.10 Active-recall prompts

1. List the 6 features of perfect competition.
2. State equilibrium condition for any firm.
3. Differentiate first-, second-, third-degree price discrimination.
4. Describe the kinked demand curve in 3 lines.
5. State 3 examples each for PC, monopolistic competition, oligopoly, monopoly.

CHAPTER 12 — Indian Economy

Importance: ★★★★★ CRITICAL (≈8 Qs / paper) **Difficulty:** Easy-Medium. Heavy on facts and current-affairs-adjacent data. Memorise the depth tables.

12.0 — Understanding the Indian Economy from First Principles

QUICK RECALL

How do we measure a whole economy's output — and why does it matter for policy?

GDP (Gross Domestic Product) is the total market value of all final goods and services produced within a country's borders in a year. "Final" goods exclude intermediate inputs — the steel that goes into a car is not counted separately (only the car is), to avoid double-counting.

Three approaches to measuring GDP (all give the same answer in theory): 1. **Production/Output approach:** Sum value added by each producer. Value added = Output value – Input value (avoids double-counting). 2. **Income approach:** Sum all incomes earned: wages + profit + rent + interest. Every rupee spent on output becomes someone's income. 3. **Expenditure approach:** $GDP = C + I + G + (X - M)$. Consumer spending + Investment + Government spending + Net exports.

Why the structural mismatch matters: Agriculture contributes ≈15–17% of GDP but employs ≈40–45% of India's workforce. Industry contributes ≈28% with far fewer workers. Services contributes ≈55–57% with educated, urban workers. This mismatch — too many people competing for a small share of GDP in agriculture — is the structural cause of rural poverty and the urban-rural income gap.

National income aggregates — the chain of adjustments:

$$GNP = GDP + NFIA \text{ (Net Factor Income from Abroad)}$$

$$NNP_{MP} = GNP - \text{Depreciation (CCA)}$$

$$NNP_{FC} = \text{National Income} = NNP_{MP} - \text{Net Indirect Taxes}$$

(Net Indirect Taxes = Indirect Taxes – Subsidies)

$$\text{Per Capita Income} = \frac{\text{National Income}}{\text{Population}}$$

Term	Concept
GDP	Production within borders (includes output by foreigners in India; excludes Indians abroad)
GNP	Production by nationals (includes Indians abroad; excludes foreigners in India)
NNP	GNP minus depreciation of capital (Net = after accounting for capital wear-out)
NI at FC	NNP at MP minus net indirect taxes (removes tax distortions; reflects factor incomes)
Real vs Nominal	Real GDP adjusts for inflation; Nominal does not. Real GDP growth = true economic growth
GDP Deflator	$\text{Nominal GDP} \div \text{Real GDP} \times 100$ (measures overall price changes in the economy)

Solved Example 12.A: GNP at MP = ₹100 lakh crore. Depreciation = ₹8 lakh crore. Indirect taxes = ₹15 lakh crore. Subsidies = ₹5 lakh crore. Find National Income.

$NNP_{MP} = 100 - 8 = 92$ lakh crore. Net indirect taxes = $15 - 5 = 10$ lakh crore. $NI = NNP_{FC} = 92 - 10 = 82$ lakh crore. **Answer: ₹82 lakh crore.**

12.1 Examiner mindset

Angle	Pet question
Nature of Indian economy	Mixed economy (public + private)
Sectoral share in GDP & employment	Service > Industry > Agriculture (in GDP); Agriculture employs the most
National income concepts (GDP, GNP, NDP, NNP)	Direct definition
Methods of measuring NI	Production, Income, Expenditure
Body that measures NI	NSO (since 2019; earlier CSO)
Population — census, growth rate	Provisional 2021 / 2011 census facts
Poverty estimates	Tendulkar, Rangarajan
Unemployment	Types — structural, cyclical, frictional, disguised
Infrastructure	Energy, transport, communication

12.2 Nature of Indian economy

India has a **mixed economy** — coexistence of public and private sectors, planning + market mechanisms. Article 39(b) and (c) of the Constitution direct distribution of resources for common good.

Sectoral classification

Sector	Activities
Primary	Agriculture, fishing, mining, forestry — direct from natural resources
Secondary	Manufacturing, construction, electricity — converts raw to finished
Tertiary	Services — banking, insurance, transport, IT, healthcare, education
Quaternary	Knowledge sector — R&D, IT, consultancy
Quinary	Highest-level decision-making — top govt + corporate executives

Sectoral share in India (recent estimates, %)

Sector	Share in GVA	Share in employment
Agriculture & allied	~17 %	~45 %
Industry	~28 %	~25 %
Services	~55 %	~30 %

Key trend: Services sector dominates GDP but **agriculture employs the largest share** of workforce — a sign of underemployment / disguised unemployment.

12.3 National Income concepts

Concept	Formula
GDP (Gross Domestic Product) at MP	Total market value of all final goods/services produced within country in a year, at current market prices
GDP at FC	GDP at MP – Indirect Taxes + Subsidies
GNP at MP	GDP at MP + Net Factor Income from Abroad (NFIA)
NDP at MP	GDP at MP – Depreciation
NNP at MP	GNP at MP – Depreciation
NNP at FC = National Income	NNP at MP – Indirect Taxes + Subsidies
Per Capita Income (PCI)	National Income / Population
Personal Income	NI – Corporate tax – Undistributed profits – Social security contributions + Transfer payments
Disposable Income	Personal Income – Personal taxes

Memorise the chain: GDP → (add NFIA) → GNP → (less Depreciation) → NNP → (less IT + Subsidies) → **National Income (NNPFC)**.

Methods of measuring National Income

Method	What it sums	Body responsible
Production / Value-added	Sum of value added by each sector	NSO
Income method	Sum of incomes (rent, wages, interest, profit)	NSO
Expenditure method	$C + I + G + (X - M)$	NSO

Body: National Statistical Office (NSO) — formed in 2019 by merging CSO + NSSO under MoSPI.

12.4 Population

Item	Detail
Latest census	2011 (next was due 2021, postponed due to COVID)
2011 population	121.08 crore (about 1.21 billion)
Decadal growth (2001-11)	17.7 %
Sex ratio (2011)	943 females per 1000 males
Child sex ratio (0-6)	919
Literacy rate (2011)	74.0 % (Male 82.1 %; Female 65.5 %)
Most populous state	Uttar Pradesh
Most densely populated state	Bihar
Largest area-wise state	Rajasthan

Demographic dividend

India has a young population (median age ~28 years), giving a "demographic dividend" — a potential boost to economic growth from a higher working-age population. Window: 2018-2050.

12.5 Poverty

Committee	Year	Methodology
Suresh Tendulkar Committee	2009	Used MPCE (Monthly Per Capita Consumption Expenditure); poverty line: ₹816 (rural) and ₹1000 (urban) per capita per month (2011-12)
C. Rangarajan Committee	2014	Higher cut-off: ₹972 (rural) and ₹1407 (urban) per month
Multidimensional Poverty Index (MPI)	UNDP / NITI Aayog	Health, education, standard of living dimensions

Tendulkar (2011-12)	Rangarajan (2011-12)
21.9 % below poverty	29.5 % below poverty

NITI Aayog's national MPI report (2021) put MPI at 0.118; about **25 %** of population multidimensionally poor in 2015-16.

12.6 Unemployment — types

Type	Definition	Example
Structural	Mismatch between skills and available jobs	Coal worker after mine closure
Cyclical	Caused by downturns in business cycle	Recession of 2008
Frictional	Short-term, between jobs	New graduate searching
Seasonal	Off-season idle	Sugarcane labour after harvest
Disguised	More people employed than needed; MP = 0	Indian agriculture
Open	Willing & able but no job	Educated youth without job
Underemployment	Employed less than capacity / less productively	Engineer driving cab

12.7 Infrastructure

Energy mix (India, recent)

Source	Share in installed capacity (%)
Coal & lignite	~50 %
Renewable (Solar + Wind + Hydro + Bio)	~43 %
Gas	~5 %
Nuclear	~2 %

India target: **500 GW non-fossil capacity by 2030** (Net-Zero by 2070, COP-26 commitment).

Transport

Mode	Length
Roads	~63 lakh km (largest in world after USA)
Railways	~68,000 route km (4th largest network)
Waterways	14,500 km navigable; National Waterways : 111 declared
Airports	148 operational
Ports	13 major + 200+ minor/intermediate

Communication

Item	Detail
Total telephone subscribers	~120 crore (90 % wireless)
Internet users	~85 crore
Telephone density	~85 %

12.8 Worked PYQ-style examples

Q 12.1. $GNP = GDP + ?$

Net Factor Income from Abroad (NFIA).

Q 12.2. National Income (NNPFC) = ?

NNP at MP – Indirect Taxes + Subsidies.

Q 12.3. Body responsible for measuring NI in India:

NSO (under MoSPI; merger of CSO and NSSO in 2019).

Q 12.4. Largest sector in India's GDP:

Services (~55 %).

Q 12.5. Largest sector in India's employment:

Agriculture (~45 %).

Q 12.6. Tendulkar Committee was on:

Poverty estimation (2009).

Q 12.7. Census 2011 population:

~121 crore.

Q 12.8. Disguised unemployment is most prevalent in:

Agriculture in India.

Q 12.9. Per Capita Income = ?

National Income / Population.

Q 12.10. Structural unemployment is caused by:

Skill / location mismatch (e.g., loss of jobs due to technology).

Q 12.11. GDP deflator = ?

Nominal GDP / Real GDP $\times 100$.

Q 12.12. A measure of standard of living:

Per Capita Income (or HDI for broader).

Q 12.13. " $C + I + G + (X - M)$ " is which method of NI?

Expenditure method.

Q 12.14. Most populous state of India?

Uttar Pradesh.

Q 12.15. Rangarajan Committee poverty cut-off (rural, monthly per capita)?

₹972 (vs Tendulkar's ₹816).

12.9 Trap-recognition card

Trap	Defence
Agriculture biggest in GDP	No — Services is biggest in GDP; agriculture biggest in employment.
GDP includes NFIA	No — that's GNP.
NI = GDP at MP	No — NI = NNP at FC.
NSO replaced FC	No — NSO replaced CSO+NSSO; nothing to do with FC.
Tendulkar = 2014	No — 2009. Rangarajan = 2014.

12.10 Mini-mock

#	Q	Answer
1	NI in India calculated by?	NSO
2	Methods of NI?	Production, Income, Expenditure
3	Largest sector in India's employment?	Agriculture
4	Tendulkar Committee year?	2009
5	Census 2011 population?	121 crore
6	Sex ratio 2011?	943
7	Disguised unemployment found in?	Agriculture
8	National Income = ?	NNPFC

#	Q	Answer
9	GDP deflator formula?	Nominal/Real × 100
10	Net-Zero target year?	2070

12.11 Active-recall prompts

1. State the chain GDP → NI with all steps.
2. Name the 3 methods of NI calculation and the body responsible.
3. List 5 types of unemployment with one example each.
4. Compare Tendulkar vs Rangarajan poverty estimates.
5. State sectoral share in India's GDP and employment.

CHAPTER 13 — Economic Reforms in India (1991 onwards)

Importance: ★★★★★ HIGH (≈4 Qs / paper) **Difficulty:** Easy. Pure factual.

13.0 — Understanding the 1991 Reforms from First Principles

QUICK RECALL

Why did India need economic reforms in 1991? What was the crisis?

By May–June 1991, India's foreign exchange reserves had fallen to barely USD 1 billion — enough for only 2 weeks of imports. India was forced to physically airlift 67 tonnes of gold to the Bank of England and 20 tonnes to the Bank of Japan as collateral for emergency loans. The government was on the brink of defaulting on its international payments — a sovereign default that would have shut India out of global capital markets.

The causes were structural and had been building for decades:

- **Licence Raj:** Every private investment required government approval ("industrial licence"). This created massive inefficiency, corruption, and protected incumbent firms from competition. An entrepreneur wanting to set up a factory often waited years for approvals.
- **Public sector dominance:** Government-owned companies controlled most key industries — they were chronically loss-making, overstaffed, and insulated from competition.
- **Fiscal profligacy:** Persistent large fiscal deficits, financed by borrowing from the RBI (printing money) and from abroad, built up unsustainable debt.
- **External shock:** The Gulf War (1990–91) caused an oil price spike (India imported most of its oil) AND disrupted remittances from Indian workers in Gulf countries — a double hit to the balance of payments.

The response — the LPG reforms (1991):

- Architect: Finance Minister **Dr. Manmohan Singh**, under PM **P.V. Narasimha Rao**.
- India negotiated a USD 2.2 billion loan from the IMF, conditioned on structural reform.

LPG — what each pillar actually changed:

Pillar	What changed	Key examples
Liberalisation	Removed government controls; dismantled Licence Raj	Industrial Licensing Policy abolished (except 6 industries); import duties slashed; FDI opened
Privatisation / Disinvestment	Reduced state role in industry; sold government stakes in PSUs	Disinvestment Commission set up; BALCO, Maruti stake sold
Globalisation	Opened economy to foreign trade and capital	Rupee made convertible on current account; FII investment allowed in stock markets; WTO membership

Results of 1991 reforms (asked frequently):

GDP growth rate rose: 5–6% in 1980s → 6–8% in 1990s–2000s → 7–8%+ in 2000s

Foreign exchange reserves rose from near-zero (1991) to USD 700+ billion (2024)

IT and services sector boom (BPO, software exports) — enabled by telecom liberalisation

Poverty rate fell significantly (though debate continues on methods)

Inequality increased (Gini coefficient rose); urban-rural gap widened

13.1 Examiner mindset

Angle	Pet question
Year of reforms	1991 (under Manmohan Singh as FM, P.V. Narasimha Rao PM)
LPG full forms	Liberalisation, Privatisation, Globalisation
Disinvestment / privatisation	Differences
Key acts of reform period	FERA → FEMA (1999), MRTP → Competition Act (2002)
Impact on growth, FDI, services	Standard impact list

13.2 Background of 1991 crisis

Trigger	Detail
Foreign exchange crisis	Reserves down to 2 weeks of imports (₹1.2 billion); needed IMF bailout
Fiscal deficit	~8.5 % of GDP
Inflation	~17 %
Gold pledged	Government pledged 67 tonnes of gold to Bank of England & Switzerland
IMF + World Bank	Provided structural adjustment loan with conditionalities → reforms
Architects	PM Narasimha Rao + FM Manmohan Singh + RBI Gov C. Rangarajan
New Industrial Policy	24 July 1991

13.3 LPG — Liberalisation, Privatisation, Globalisation

Liberalisation

*Removing **government controls** on the economy to allow markets to function freely.*

Area	Reform
Industrial licensing	Abolished for all industries except 6 (now 4) — alcohol, defence, atomic energy, hazardous chemicals (industrial explosives & defence/atomic remain reserved or licensed)
De-reservation	Most areas earlier reserved for public sector opened to private
Small-scale reservation	Progressively phased out
M RTP Act	Replaced by Competition Act 2002 (CCI established)
FERA → FEMA	Foreign Exchange Regulation Act 1973 replaced by Foreign Exchange Management Act 1999
Tax reforms	Lower rates, broader base; introduction of GST in 2017
Banking	Entry of private/foreign banks; Narasimham Committee reforms

Privatisation

Transfer of ownership / management from public sector to private sector.

Mode	Description
Disinvestment	Sale of part of govt's equity in PSU (govt may still retain control)
Strategic disinvestment	Govt transfers > 50 % equity + management control to private
Outright privatisation	Full sale (rare in India before recent years)

| Body | Department of Investment and Public Asset Management (**DIPAM**), Ministry of Finance |

Globalisation

Integration of national economy with the world economy.

Area	Reform
Tariff cuts	Customs duty reduced from peak >300 % to ~10 %
Convertibility	Current-account convertibility (1994); partial capital convertibility
FDI route	Automatic route for most sectors; liberal sectoral caps
WTO	India joined 1 Jan 1995 (founding member)
Trade share	India's share in global trade rose from 0.5 % (1990) to ~2 % (now)

13.4 Major institutional changes post-1991

Institution	Change
SEBI	Statutory body in 1992 (SEBI Act) — securities market regulator
Competition Commission of India (CCI)	2003 (Act 2002) — replaced MRTP
TRAI	1997 — telecom regulator
IRDA	1999 — insurance regulator (now IRDAI)
NITI Aayog	2015 — replaced Planning Commission
GST Council	2017 — under Art. 279A

13.5 Worked PYQ-style examples

Q 13.1. New Industrial Policy was announced in:

24 July 1991.

Q 13.2. LPG stands for:

Liberalisation, Privatisation, Globalisation.

Q 13.3. FEMA replaced FERA in:

1999.

Q 13.4. Competition Commission of India (CCI) was set up under:

Competition Act 2002 (operational 2003).

Q 13.5. GST was introduced in India on:

1 July 2017.

Q 13.6. Body that handles disinvestment in India:

DIPAM (Dept. of Investment and Public Asset Management).

Q 13.7. Number of industries reserved for public sector after 1991 reforms (currently):

2 (atomic energy and railway operations) — most sectors opened up; railway operations partially opened to private.

Q 13.8. SEBI was given statutory status in:

1992 (SEBI Act).

Q 13.9. WTO was joined by India on:

1 January 1995 (founding member).

Q 13.10. Strategic disinvestment means:

Sale of > 50 % equity + transfer of management control to private buyer.

13.6 Trap-recognition card

Trap	Defence
LPG = Liquefied Petroleum Gas	In economics context: Liberalisation, Privatisation, Globalisation
FERA replaced FEMA	Wrong direction — FEMA replaced FERA
MRTTP still operative	Replaced by Competition Act 2002
Disinvestment = privatisation	Disinvestment is partial; strategic disinvestment $\approx$ privatisation
GST in 2016	No — 1 July 2017

13.7 Mini-mock

#	Q	Answer
1	Year of reforms	1991
2	FM during 1991 reforms	Manmohan Singh
3	PM during 1991 reforms	P.V. Narasimha Rao
4	LPG full form	Liberalisation, Privatisation, Globalisation
5	FEMA year	1999
6	GST year	2017
7	NITI Aayog year	2015
8	DIPAM stands for	Dept. of Investment and Public Asset Management
9	WTO joined India	1 Jan 1995
10	SEBI statutory status	1992

13.8 Active-recall prompts

1. List 5 elements of liberalisation.
2. Differentiate disinvestment vs strategic disinvestment.
3. State 5 institutions established / strengthened post-1991.
4. Why was the 1991 BoP crisis triggered?
5. What does "automatic route" of FDI mean?

CHAPTER 14 — Money & Banking (RBI, Commercial Banks, Monetary Policy)

Importance: ★★★★★ CRITICAL (≈7 Qs / paper) **Difficulty:** Medium. Heavy on facts (RBI tools, governors, dates).

14.0 — Understanding Money & Banking from First Principles

QUICK RECALL

What IS money — and how do banks create it from thin air?

Money is not just physical cash. Economists define it by its three functions: 1. **Medium of exchange:** Eliminates the barter problem (you don't need someone who has what you want AND wants what you have). 2. **Store of value:** You can earn money today and spend it next month. It holds value over time (subject to inflation eroding it). 3. **Unit of account:** Prices are denominated in rupees — a common measuring stick for all goods.

Historically: cowrie shells → gold coins → paper notes backed by gold → modern fiat money (backed by nothing except trust in the government and the central bank). India's rupee is **fiat money** — the RBI declares it legal tender; its value comes from government authority and public trust.

How commercial banks create money — the multiplier effect: You deposit ₹1,000 in Bank A. The RBI requires Bank A to keep 4% as CRR (Cash Reserve Ratio) = ₹40. Bank A can lend ₹960 to someone. That person spends ₹960 → it gets deposited in Bank B. Bank B keeps ₹38.4 (4% CRR) and lends ₹921.6. This chain continues. Total money created = $\frac{₹1,000}{CRR} = \frac{1,000}{0.04} = ₹25,000$.

Money Multiplier = $1/CRR$. A higher CRR means a smaller multiplier — the RBI can contract money supply by raising CRR without printing or destroying any physical currency.

RBI's monetary policy tools — how the central bank controls money:

Tool	Mechanism	Raising it →	Lowering it →
Repo Rate	Rate at which RBI lends to commercial banks	Banks borrow less → less lending → money supply ↓ → inflation ↓	Banks borrow more → more lending → money supply ↑ → growth ↑
Reverse Repo Rate	Rate at which RBI borrows from commercial banks	Banks park more with RBI → less lent to public → credit tightens	Banks prefer to lend to public → credit expands
CRR (Cash Reserve Ratio)	% of deposits kept as cash with RBI	Money multiplier falls → money supply contracts	Money multiplier rises → money supply expands
SLR (Statutory Liquidity Ratio)	% of deposits kept in liquid assets (Govt securities, gold)	Less funds for loans → credit contraction	More funds for loans → credit expansion
Bank Rate	Rate at which RBI rediscounts bills of commercial banks	Signals tight policy; banks raise lending rates	Signals loose policy
OMO (Open Market Operations)	RBI buys/sells govt securities in open market	Selling: absorbs liquidity from market	Buying: injects liquidity into market
MSF (Marginal Standing Facility)	Emergency overnight borrowing window for banks	—	Banks can borrow up to 1% of NDTL above SLR

Money supply aggregates (M0 to M4) — broader = less liquid:

Aggregate	Components	Nickname
M0	Currency in circulation + Bankers' deposits with RBI + Other deposits with RBI	Reserve Money / High-powered money
M1	Currency with public + Demand deposits with banks	Narrow Money
M2	M1 + Savings deposits with Post Office (excluding NSC)	—
M3	M1 + Time deposits with banks	Broad Money (most used for policy)
M4	M3 + All Post Office deposits	Broadest measure

Solved Example 14.A: CRR = 4%. A bank receives a fresh deposit of ₹50,000. What is the maximum credit creation?

- *Money Multiplier* = $1/CRR = 1/0.04 = 25$
- *Credit creation* = $\text{Deposit} \times \text{Money Multiplier} = ₹50,000 \times 25 = ₹12,50,000$

14.1 Examiner mindset

Angle	Pet question
Functions of money	Medium of exchange / Measure of value / Store of value / Standard of deferred payment
Money supply measures	M0, M1, M2, M3, M4
RBI estd date	1 April 1935; nationalised 1949
Functions of RBI	6 traditional + 5 developmental
Quantitative tools	Bank rate, Repo, Reverse repo, CRR, SLR, OMO, MSF, Standing Deposit Facility

Angle	Pet question
Qualitative tools	Margin, moral suasion, direct action, credit rationing
Commercial bank functions	Primary + Secondary
Types of banks	Scheduled vs non-scheduled, public vs private, RRB, payment, small finance
Monetary Policy Committee (MPC)	6 members; majority view; RBI Governor casting vote

14.2 Money — definitions

Money is anything that is generally accepted as a medium of exchange, measure of value, store of value, and standard of deferred payment.

Functions

Function	Detail
Primary	Medium of exchange + Measure of value
Secondary	Store of value + Standard of deferred payment + Means of transfer of value
Contingent	Distribution of national income, basis of credit, liquidity

Money supply (RBI's measures)

Aggregate	Components
M0 (Reserve Money)	Currency in circulation + Bankers' deposits with RBI + Other deposits with RBI
M1 (Narrow Money)	Currency with public + Demand deposits with banks + Other deposits with RBI
M2	M1 + Savings deposits with Post Office
M3 (Broad Money)	M1 + Time deposits with banks (most commonly used aggregate)
M4	M3 + Total deposits with Post Office Savings Bank (excluding NSC)

Order of liquidity: M1 > M2 > M3 > M4 (M1 most liquid).

14.3 Reserve Bank of India

Item	Detail
Established	1 April 1935 (under RBI Act 1934)
First Governor	Sir Osborne Smith
First Indian Governor	C.D. Deshmukh
Nationalised	1 January 1949
Headquarters	Mumbai (originally Calcutta)
Current Governor	Sanjay Malhotra (took office 11 Dec 2024; succeeded Shaktikanta Das)
Initial paid-up capital	₹5 crore

Functions of RBI

(A) Traditional (Central banking) functions

- Issue of currency** (sole authority for notes $\geq$ ₹2; ₹1 notes & all coins issued by Gol but circulated by RBI)
- Banker to government** (centre + states; manages public debt)
- Banker's bank** (CRR holding; lender of last resort)
- Custodian of foreign exchange reserves**
- Controller of credit / monetary authority**
- Lender of last resort**

(B) Developmental & promotional

- Develops banking infrastructure (RRBs, NABARD, NHB)
- Provides agricultural finance (via NABARD, since 1982)
- Industrial finance (via IDBI earlier)
- Promotes financial inclusion (PMJDY, payment banks)
- Regulates and supervises banks (BR Act 1949)

(C) Supervisory

- Inspection of banks

2. Licensing
3. Approval of mergers, branches

14.4 Monetary Policy Tools

Quantitative tools (affect overall money supply)

Tool	Description	Current rate (illustrative; check at exam time)
Bank Rate	Rate at which RBI lends to commercial banks (long-term, without collateral)	6.75 %
Repo Rate	Rate at which RBI lends to banks against govt securities (short-term)	6.50 % (key policy rate)
Reverse Repo Rate	Rate at which RBI borrows from banks (absorb liquidity)	3.35 % (effectively replaced by SDF)
Standing Deposit Facility (SDF)	New tool (2022); banks park surplus liquidity at this rate	6.25 %
MSF (Marginal Standing Facility)	Emergency overnight lending to banks against SLR securities	6.75 %
CRR (Cash Reserve Ratio)	% of NDTL banks must keep with RBI as cash	4.50 %
SLR (Statutory Liquidity Ratio)	% of NDTL banks must keep in approved liquid assets (gold, govt securities)	18.00 %
OMO (Open Market Operations)	RBI buys/sells govt securities to inject/absorb liquidity	continuous

Qualitative tools (selective credit control)

Tool	Description
Margin requirements	Lower margin → easier credit
Moral suasion	Persuasion / advice to banks
Credit rationing	Limits on credit to specific sectors
Direct action	Penalties for non-compliance
Consumer credit regulation	Down-payment, EMI tenure norms
Publicity	Influencing public expectations

Monetary Policy Committee (MPC)

Item	Detail
Established	2016 (RBI Act amendment)
Members	6 (3 RBI + 3 govt-nominated)
Chair	RBI Governor
Decision	Majority vote; Governor has casting vote in tie
Meets	At least 4 times a year
Inflation target	4 % ±2 % (CPI-Combined; legal mandate till March 2026)

14.5 Commercial Banks

Functions

(A) Primary functions

1. **Accepting deposits** - Demand: current, savings (chequable, instantly withdrawable) - Time: fixed, recurring (longer tenure, higher interest)
2. **Lending money** - Loans, cash credit, overdraft, bill discounting

(B) Secondary / Agency functions

Type	Examples
Agency	Collecting cheques/bills, paying utility bills, sale-purchase of securities
General utility	Locker, foreign exchange, letter of credit, gift cheques, ATM service

Process of credit creation

*Commercial banks create credit/deposits through the **money multiplier** = $1/CRR$ (in simplified model).*

If primary deposit = ₹1000 and CRR = 10 %, banks can create deposits up to ₹10,000.

Types of banks in India

Type	Examples
Scheduled Commercial Banks (SCBs)	Listed in 2nd Schedule of RBI Act
Public Sector Banks (PSBs)	SBI + 11 nationalised banks (after consolidation)
Private Banks	HDFC, ICICI, Axis, Kotak, etc.
Foreign Banks	Citibank, Standard Chartered, HSBC, etc.
Regional Rural Banks (RRBs)	Established 1975; sponsored by PSBs; serve rural areas
Cooperative Banks	Urban + rural cooperative
Payment Banks	(since 2015) — accept deposits up to ₹2 lakh, no lending; e.g., Airtel, Paytm, India Post
Small Finance Banks (SFBs)	(since 2015) — focus on small businesses, MSMEs, low-income; e.g., Equitas, Ujjivan
Local Area Banks	Limited to 3 contiguous districts

Major banking committees

Committee	Year	Subject
Hilton Young Commission	1926	Recommended creation of RBI
Narasimham Committee I	1991	First-generation banking reforms (interest deregulation, capital adequacy)
Narasimham Committee II	1998	Second-generation reforms
Raghuram Rajan Committee	2008	Financial sector reforms
Bimal Jalan Committee	2014	New bank licences
Nachiket Mor Committee	2014	Recommended payment & small finance banks
Urjit Patel Committee	2014	Inflation targeting framework (led to MPC)

14.6 Worked PYQ-style examples

Q 14.1. Sole authority to issue currency notes in India:

RBI (under Section 22 of RBI Act, 1934) for notes $\geq ₹2$.

Q 14.2. First Governor of RBI:

Sir Osborne Smith.

Q 14.3. RBI was nationalised in:

1 January 1949.

Q 14.4. Most liquid measure of money supply:

M1.

Q 14.5. Reverse repo rate is the rate at which:

RBI **borrow**s from banks (absorbs liquidity).

Q 14.6. SDF stands for:

Standing Deposit Facility (introduced April 2022).

Q 14.7. Inflation target of MPC:

4 % $\pm$ 2 % CPI-Combined.

Q 14.8. MPC has how many members?

6 (3 RBI + 3 govt-nominated).

Q 14.9. Banker to the Government in India:

RBI.

Q 14.10. Money multiplier in simple model:

1 / CRR.

Q 14.11. RRB sponsored by:

Sponsor bank (a PSB) + GoI + State Govt (50:35:15).

Q 14.12. Payment banks were introduced based on:

Nachiket Mor Committee.

Q 14.13. Bank rate vs Repo rate — which is collateralised?

Repo (against govt securities); Bank Rate is uncollateralised.

Q 14.14. Open Market Operations (OMO) involve RBI:

Buying / selling government securities in the open market.

Q 14.15. Inflation targeting was introduced in India in:

2016 (after Urjit Patel Committee).

14.7 Trap-recognition card

Trap	Defence
RBI established 1949	No — established 1935; nationalised 1949
Bank rate is collateralised	No — that's repo. Bank rate is uncollateralised
MPC has 12 members	No — 6
CRR is held in govt securities	No — held as cash with RBI. SLR is in govt securities + gold
RRBs are private banks	No — public-sector ownership; sponsored by PSBs
Inflation target = 6 %	No — 4 % is the target; 6 % is the upper tolerance band

14.8 Mini-mock

#	Q	Answer
1	RBI estd	1 April 1935
2	RBI nationalised	1949
3	First Governor	Osborne Smith
4	Headquarters	Mumbai
5	Sole currency issuer	RBI
6	M3 = ?	M1 + Time deposits
7	Inflation target	4 % $\pm$ 2 %

#	Q	Answer
8	MPC members	6
9	SDF year	2022
10	CRR currently	~4.50 %

14.9 Active-recall prompts

1. List 6 traditional functions of RBI.
2. State the difference between repo and bank rate.
3. Define M0, M1, M3.
4. Name 3 banking committees and their contributions.
5. Explain the role of MPC.

CHAPTER 15 — Fiscal Policy, Budget, FRBM Act & Balance of Payments

Importance: ★★★★★ CRITICAL (≈7 Qs / paper) **Difficulty:** Medium. Definition + computation. Always 2 questions on deficits.

15.0 — Understanding Fiscal Policy & Budget from First Principles

QUICK RECALL

What is fiscal policy and why does the government's deficit matter?

While the RBI manages **monetary policy** (money supply and interest rates), the government manages **fiscal policy** — its decisions about spending and taxation. Fiscal policy affects the economy through the multiplier: when the government spends ₹100 on roads, road construction workers get income, they spend it on food and clothes, those sellers earn income, they spend it — each rupee circulates, creating more than ₹100 of total demand. This is the Keynesian fiscal multiplier.

When expenditure > revenue → the government must borrow. The amount borrowed is the **fiscal deficit**. This is not automatically bad — borrowing to build productive infrastructure (roads, ports, power) can generate returns that exceed the interest cost. But borrowing to cover routine expenses (subsidies, salaries when revenue is insufficient) creates a debt spiral: interest on past debt consumes more and more of future revenue, leaving less for development.

The three deficit concepts you must distinguish: - **Fiscal Deficit (FD):** Total government expenditure – (Revenue receipts + Non-debt capital receipts). This equals the government's total borrowing requirement for the year. - **Revenue Deficit (RD):** Revenue expenditure – Revenue receipts. Shows whether the government is borrowing just to cover current expenses (very bad) or for capital investment (less bad). - **Primary Deficit (PD):** Fiscal Deficit – Interest payments. Measures the "new" borrowing need, excluding debt servicing on past loans. If PD = 0, new borrowing only covers interest on old debt — the debt stock is stable.

FORMULA

The three deficit formulas (memorise all three):

$$\text{Fiscal Deficit} = \text{Total Expenditure} - (\text{Revenue Receipts} + \text{Non-debt Capital Receipts})$$

$$\text{Revenue Deficit} = \text{Revenue Expenditure} - \text{Revenue Receipts}$$

$$\text{Primary Deficit} = \text{Fiscal Deficit} - \text{Interest Payments}$$

$$\text{Monetised Deficit (formerly)} = \text{amount borrowed from RBI (money printing)}$$

Budget classification — the two-part budget:

Budget	Receipts	Expenditure
Revenue budget	Tax + non-tax revenues; recurring, do NOT create assets	Salaries, subsidies, interest payments, grants; recurring, do NOT create assets
Capital budget	Borrowings, disinvestment, recoveries of loans; create obligations	Infrastructure spending, loans to states, repayment of old loans; create assets

FRBM Act (Fiscal Responsibility and Budget Management Act, 2003):

Aims to eliminate **revenue deficit** and reduce **fiscal deficit** to 3% of GDP.

Requires the government to present a Fiscal Policy Strategy Statement with the budget.

Has been amended multiple times — targets relaxed post-COVID (India's FD was ≈ 6.4% in FY21). Currently converging back toward 4.9% (FY25 revised estimate).

Balance of Payments (BoP):

Account	Records	Examples
Current Account	Trade in goods + services + transfers	Exports, imports, remittances, interest payments
Capital Account	Small capital transfers	Debt write-offs, sale of non-financial assets
Financial Account	Investment flows	FDI, FII, external borrowings, forex reserve changes

BoP always balances to zero by definition (any current account deficit is financed by a capital/financial account surplus or a fall in reserves).

Solved Example 15.A: FD = ₹16 lakh crore. Interest payments = ₹11 lakh crore. Revenue receipts = ₹26 lakh crore. Revenue expenditure = ₹37 lakh crore.

$$\text{Revenue Deficit} = 37 - 26 = \text{₹11 lakh crore} \quad \text{Primary Deficit} = \text{FD} - \text{Interest} = 16 - 11 = \text{₹5 lakh crore}$$

15.1 Examiner mindset

Angle	Pot question
Fiscal policy tools	Taxation, public spending, public debt
Budget components	Revenue Receipts/Expenditure, Capital Receipts/Expenditure
Types of deficits	Revenue, Fiscal, Primary, Effective Revenue
FRBM Act 2003	Targets, exit clauses
Balance of Payments components	Current Account, Capital Account

Angle	Pet question
BoP imbalance	Current account deficit / surplus
Direct vs Indirect tax	Examples and incidence
Major taxes after GST	Direct (income, corporate); Indirect (GST, customs, excise on petroleum/alcohol)

15.2 Fiscal Policy

Fiscal policy = government policy regarding **taxation**, **public expenditure**, and **public debt** to achieve macroeconomic objectives (growth, stability, employment, equity).

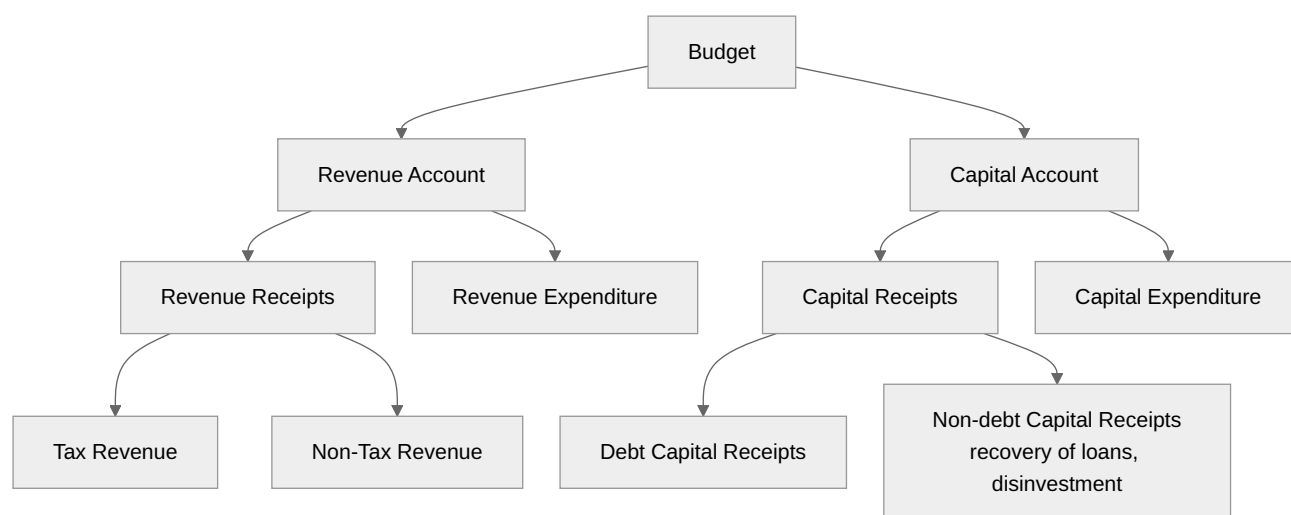
Tool	Effect
Taxation	Reduces disposable income; controls demand
Public expenditure	Direct demand creation; multiplier effect
Public borrowing	Mobilises savings; can crowd out private investment
Subsidies & transfers	Income support to vulnerable groups

15.3 Budget — structure

Annual Financial Statement is the technical name for the Union Budget under Article 112 of the Constitution.

Article	Provision
Art. 112	Annual Financial Statement
Art. 113	Procedure of estimates in Parliament
Art. 114	Appropriation Bills
Art. 110	Definition of Money Bill
Art. 266	Consolidated Fund and Public Account
Art. 267	Contingency Fund

Components of Budget



Receipts

Type	Examples
Revenue Receipts	Tax (Direct + Indirect) + Non-tax (interest, dividend, fees) — neither create liability nor reduce asset
Capital Receipts	Loans (debt) + Disinvestment + Recovery of loans (non-debt) — create liability or reduce assets

Expenditure

Type	Examples
Revenue Expenditure	Salaries, subsidies, interest payments, defence services revenue, social services

Type	Examples
Capital Expenditure	Asset-creating: roads, bridges, equipment, investments, loans by Govt

Plan vs Non-plan (terminology dropped 2017)

Earlier: Plan = part of Five-Year Plan; Non-plan = recurring expenditure. Now: replaced by **Capital vs Revenue** classification only.

15.4 Three Funds

Fund	Article	Description
Consolidated Fund of India	Art. 266(1)	All revenue, loan receipts, loan recoveries — withdrawal needs Parliament's authority
Public Account of India	Art. 266(2)	Money held in trust (PF, small savings) — withdrawal doesn't need Parliament's approval
Contingency Fund of India	Art. 267	₹30,000 crore corpus (raised 2021) — for unforeseen expenditure; at President's disposal; recoupable from CFI

15.5 Types of Deficits

Deficit	Formula
Revenue Deficit	Revenue Expenditure – Revenue Receipts
Effective Revenue Deficit	Revenue Deficit – Grants for creation of capital assets
Budget Deficit	Total Expenditure – Total Receipts (now negligible after rule change)
Fiscal Deficit	Total Expenditure – (Revenue Receipts + Non-debt Capital Receipts) = govt's borrowing requirement
Primary Deficit	Fiscal Deficit – Interest payments
Monetised Deficit	Borrowing from RBI (now restricted by FRBM)

Memory hook: Fiscal deficit = how much the government **borrow**s in a year (so it equals the increase in public debt, ignoring repayment).

15.6 FRBM Act, 2003

Item	Detail
Full form	Fiscal Responsibility and Budget Management Act
Year	2003 (came into force 5 July 2004)
Original target	Fiscal deficit $\leq$ 3 % of GDP; revenue deficit eliminated by FY09
Amendments	Multiple — N.K. Singh Committee (2017) recommended 3 % FD by 2020 + Debt-to-GDP target of 60 % (40 % Centre + 20 % States)
Escape clauses	National security, war, calamity, structural reforms — allow ± 0.5 % deviation
Suspended in	COVID-19 pandemic (2020-21)
Latest target	Fiscal deficit < 4.5 % of GDP by FY26 (as per Budget)

15.7 Balance of Payments

BoP = systematic record of all economic transactions between residents of a country and the rest of the world over a period.

Two main accounts

Account	What it records
Current Account	Trade in goods (visibles) + Trade in services (invisibles) + Income (interest, dividend) + Current transfers (remittances, gifts)
Capital Account	Investment flows (FDI + FPI), loans, banking capital, bank deposits, RBI's reserves change

Current Account components (in depth)

Item	Detail
Exports of goods	Visibles credit
Imports of goods	Visibles debit
Trade balance = Exports – Imports of goods	India typically has trade deficit
Services (software exports, tourism, etc.)	India has services surplus
Income (interest received vs paid, profits abroad)	Usually deficit

Item	Detail
Transfers (remittances)	India is a large recipient (~USD 110 bn/year — top in world)
Current Account Balance	Trade balance + Services + Income + Transfers

Capital Account components

Item	Detail
FDI (Foreign Direct Investment)	Long-term, direct stake ($\geq 10\%$)
FPI (Foreign Portfolio Investment)	Stocks, bonds — short-term, "hot money"
ECBs (External Commercial Borrowings)	Loans from foreign sources
NRI deposits	NRE, NRO, FCNR accounts
Banking capital	Inter-bank flows
Reserves	RBI's forex reserves (entered with negative sign when increasing)

BoP equilibrium

BoP must always balance in accounting terms ($\text{Current} + \text{Capital} + \text{Reserves} = 0$). What we colloquially call "BoP deficit" usually means **current account deficit** financed by capital account inflows or running down reserves.

15.8 Indian Tax structure (post-GST)

Direct taxes (paid by the entity on whom levied)

Tax	Authority
Income tax (individual)	CBDT, Min. of Finance
Corporate tax	CBDT
Capital gains tax	CBDT
Securities Transaction Tax (STT)	CBDT

Indirect taxes (passed on to consumer)

Tax	Authority
GST (subsumed: Excise, Service tax, VAT, CST, entry tax, octroi, etc.)	GST Council + CBIC
Customs Duty	CBIC
Excise on petroleum products & alcohol	Gol / State
Stamp Duty	State

GST structure

Tax	Levied by	On
CGST	Central Government	Intra-state supplies
SGST	State Government	Intra-state supplies
IGST	Central Government	Inter-state supplies + imports
UTGST	UT Government	Intra-UT supplies (no legislature)

GST Council: Chaired by **Union Finance Minister** + State FMs. Decisions need **3/4ths weighted majority** (Centre 1/3, States 2/3 of votes).

GST slabs: 0 %, 5 %, 12 %, 18 %, 28 % (+ cess on demerit goods).

15.9 Worked PYQ-style examples

Q 15.1. Annual Financial Statement is provided under Article:

112.

Q 15.2. Fiscal Deficit = ?

Total Expenditure – (Revenue Receipts + Non-debt Capital Receipts).

Q 15.3. Primary deficit = ?

Fiscal Deficit – Interest payments.

Q 15.4. FRBM Act enacted in:

2003.

Q 15.5. N.K. Singh FRBM Committee debt-to-GDP target?

60 % (40 % Centre + 20 % States).

Q 15.6. Money Bill is defined under Article:

110.

Q 15.7. BoP records:

All transactions between **residents** of a country and the rest of the world.

Q 15.8. Largest contributor to India's invisibles:

Software services + remittances.

Q 15.9. GST introduced in India under which Constitutional Amendment?

101st Amendment Act, 2016.

Q 15.10. GST Council voting: Centre's share?

1/3 of total votes; States have 2/3. Decision needs **3/4ths weighted majority**.

Q 15.11. Contingency Fund of India is at the disposal of:

President of India.

Q 15.12. Withdrawal from Public Account requires Parliamentary approval?

No (it's held in trust; needs no Parliamentary nod).

Q 15.13. Disinvestment receipts are which type?

Non-debt capital receipts.

Q 15.14. Interest payment is which expenditure?

Revenue expenditure.

Q 15.15. GST is which type of tax?

Indirect, **destination-based**, value-added consumption tax.

15.10 Trap-recognition card

Trap	Defence
Fiscal deficit includes interest	Yes (it's part of expenditure). Primary excludes interest.
Budget = financial statement under Art. 110	No — Art. 112 (Annual Financial Statement). Art. 110 = Money Bill definition.
Fund of India = Public Account	No — three separate funds (CFI, PA, Contingency).
FRBM target = 60 % deficit	No — that's debt-to-GDP. Fiscal deficit target is 3 % (relaxed during pandemic).
GST has 3 slabs	No — 5 slabs (0/5/12/18/28).
Capital Account includes export of goods	No — Current Account. Capital Account is about financial flows.

15.11 Mini-mock

#	Q	Answer
1	Fiscal Deficit formula	TE – (RR + Non-debt CR)
2	Primary Deficit formula	FD – Interest
3	FRBM Act year	2003
4	Budget article	112
5	Money Bill article	110
6	Three funds in Constitution	CFI (266-1), PA (266-2), Contingency (267)
7	GST CA Amendment	101st (2016)

#	Q	Answer
8	Top remittance receiver country	India (~USD 110 bn)
9	India's main BoP weakness	Trade deficit (current account)
10	RBI lends to govt under what arrangement?	Ways and Means Advances (WMA)

15.12 Active-recall prompts

1. State 4 types of deficit with formulas.
2. Differentiate Revenue vs Capital receipts.
3. List 3 funds in the Constitution and their articles.
4. State FRBM Act targets (original and N.K. Singh).
5. List components of Current and Capital Account in BoP.

CHAPTER 16 — Role of Information Technology in Governance

Importance: ★★☆☆ MEDIUM (≈4 Qs / paper) **Difficulty:** Easy. Pure factual.

16.0 — Understanding IT in Governance from First Principles

QUICK RECALL

How does IT transform governance — from "licence-raj queues" to "click-through services"?

Traditional government service delivery: visit the office → stand in a queue → submit physical documents → wait weeks (sometimes months) → pay a bribe to speed things up → get a certificate. For rural citizens without access to government offices, many services were effectively inaccessible.

e-Governance (electronic governance) replaces this with digital processes that are: faster (minutes instead of weeks), cheaper (no physical infrastructure per transaction), more transparent (automated systems reduce human discretion = less corruption), accessible (from any internet-connected device), and auditable (digital trail is harder to tamper with than paper).

The five interaction types of e-Governance: 1. **G2C (Government to Citizen):** DigiLocker for certificates, Umang app, CSCs for rural access, income/caste/birth certificates online. 2. **G2B (Government to Business):** GST portal (tax filing), MCA21 (company registration), GeM (government procurement), DGFT portal (import-export). 3. **G2G (Government to Government):** Police-court-prison data integration, central monitoring of scheme expenditures via PFMS. 4. **G2E (Government to Employee):** HRMS for leave/payroll, NPS (National Pension System) online. 5. **C2G (Citizen to Government):** MyGov portal for suggestions, CPGRAMS for grievances.

Key e-Governance initiatives — what examiners test:

Initiative	Full form / Purpose
Aadhaar	12-digit biometric identity; enables direct benefit transfer (DBT)
JAM Trinity	Jan Dhan + Aadhaar + Mobile — eliminates middlemen in subsidy delivery
DigiLocker	Cloud storage for digital documents; eliminates need to carry physical certificates
Umang	Unified Mobile Application for New-age Governance — single app for 1,200+ government services
GeM	Government e-Marketplace — transparent procurement for all government bodies
PFMS	Public Financial Management System — real-time tracking of central scheme funds
GSTN	GST Network — IT backbone for Goods and Services Tax compliance
MCA21	Ministry of Corporate Affairs portal — company registration, filings
CPGRAMS	Centralised Public Grievance Redress and Monitoring System
NeGP	National e-Governance Plan — framework launched 2006; mandated 27 Mission Mode Projects
CSC	Common Service Centre — village-level IT service delivery points (over 5 lakh centres)
NeSDA	National e-Governance Service Delivery Assessment — ranks states on e-governance

Digital India Programme (2015): Three pillars — (1) Digital Infrastructure as a Core Utility; (2) Governance and Services on Demand; (3) Digital Empowerment of Citizens.

16.1 Examiner mindset

Angle	Pot question
e-Governance pillars	G2C, G2B, G2G, G2E
Major Indian e-Governance initiatives	Digital India, UMANG, BharatNet, MyGov, DigiLocker
IT Act 2000	Cyber laws, digital signatures
Aadhaar / UIDAI	Statutory body, Aadhaar Act 2016
Direct Benefit Transfer (DBT)	JAM trinity
GovTech (UPI, ONDC, OCEN)	India Stack

16.2 e-Governance — the four pillars

Type	Description
G2C (Government to Citizen)	Public services to citizens (e-District, passport, taxes)
G2B (Government to Business)	Business interactions (GST portal, MCA21)
G2G (Government to Government)	Inter-agency communication (e-Office, NIC infrastructure)
G2E (Government to Employees)	HR services for govt staff (e-HRMS)

16.3 Major e-Governance initiatives in India

Initiative	Launch year	Description
Digital India	1 July 2015	Umbrella programme: 9 pillars including BharatNet, e-Kranti, public Wi-Fi
National e-Governance Plan (NeGP)	2006	Predecessor; set up Mission Mode Projects (MMPs)
BharatNet	2011	Optical fibre to all gram panchayats (~2.5 lakh)
UMANG	2017	Unified Mobile Application for New-age Governance — single app for 1500+ govt services
DigiLocker	2015	Cloud storage for verified documents
MyGov	2014	Citizen engagement platform
e-Hospital, e-Sign, e-Office	various	Mission Mode Projects
Direct Benefit Transfer (DBT)	1 Jan 2013	Welfare cash transfer to bank accounts
JAM trinity	2014-15	Jan Dhan + Aadhaar + Mobile
PMJDY (Jan Dhan Yojana)	28 Aug 2014	Universal banking inclusion
APIs / India Stack	various	UPI, Aadhaar e-KYC, eSign, DigiLocker, AA
Aadhaar (UIDAI)	UIDAI estd 2009; statutory under Aadhaar Act 2016	12-digit unique ID
GST Network (GSTN)	2017	Tech backbone for GST
ONDC	2022	Open Network for Digital Commerce
OCEN	2020	Open Credit Enablement Network — flow-based credit
Aarogya Setu	2020	COVID contact tracing

16.4 IT Act 2000 (and amendments)

Aspect	Detail
Year	17 October 2000
Amendment	2008 (added cyber crime sections, intermediary liability)
Based on	UNCITRAL Model Law on Electronic Commerce
Key provisions	Legal recognition of e-records, digital signatures, cyber offences
Section 66A	Was struck down by SC in Shreya Singhal v. Union of India (2015) for being vague
CCA (Controller of Certifying Authorities)	Issues licences to digital signature certifying authorities
CERT-In	Indian Computer Emergency Response Team — cyber security

16.5 Benefits and challenges of e-Governance

Benefits

1. **Speed**, accessibility, transparency
2. **Reduced corruption** (no middlemen, audit trails)
3. **Cost savings** for citizens (less travel) and government
4. **Wider reach** (rural via mobile)
5. **Improved service delivery** (passport in days, not months)

Challenges

1. **Digital divide** (rural, elderly, disabled)
2. **Cybersecurity threats** (data breaches)
3. **Data privacy** (Digital Personal Data Protection Act 2023 enacted)
4. **Infrastructure gaps** (electricity, internet)
5. **Resistance to change** in bureaucracy

16.6 Worked PYQ-style examples

Q 16.1. "Digital India" launched on:

1 July 2015.

Q 16.2. UIDAI made statutory under:

Aadhaar Act 2016.

Q 16.3. UMANG stands for:

Unified Mobile Application for **New-age Governance**.

Q 16.4. JAM trinity = ?

Jan Dhan + Aadhaar + Mobile.

Q 16.5. IT Act came into force on:

17 October 2000.

Q 16.6. Section 66A of IT Act was struck down in:

Shreya Singhal vs Union of India (2015).

Q 16.7. CERT-In handles:

Cyber security incidents; nodal agency under MeitY.

Q 16.8. ONDC stands for:

Open Network for Digital Commerce.

Q 16.9. Aadhaar is a:

12-digit unique identification number issued by UIDAI.

Q 16.10. PMJDY launch date:

28 August 2014.

16.7 Trap-recognition card

Trap	Defence
Aadhaar Act 2009	No — Act is 2016; UIDAI estd 2009
Section 66A still valid	No — struck down in 2015
Digital India launched 2014	No — 1 July 2015
UMANG = unified payment app	No — that's UPI; UMANG is for govt services
BharatNet covers urban areas	No — gram panchayats (rural)

16.8 Mini-mock

#	Q	Answer
1	Digital India launch	1 July 2015
2	IT Act year	2000
3	UMANG full form	Unified Mobile App for New - age Governance
4	UIDAI statutory under	Aadhaar Act 2016
5	JAM trinity	Jan Dhan + Aadhaar + Mobile
6	DigiLocker year	2015
7	Section struck down	66A (2015 — Shreya Singhal)
8	CERT-In handles	Cyber security
9	India's UPI launched by	NPCI (2016)
10	Aadhaar number digits	12

16.9 Active-recall prompts

1. List 4 e-governance pillars.
2. Name 5 major Digital India initiatives.
3. State the IT Act year and the section struck down.
4. Define JAM trinity.
5. List 3 challenges of e-governance.

APPENDIX A — Master Formula & Fact Sheet (one-shot revision)

A.1 Accounting

Concept	Formula / Fact
Accounting equation	Assets = Liabilities + Capital
Expanded	Assets = Liab + Cap + Rev – Exp – Drawings
COGS	Opening stock + Net purchases + Direct exp – Closing stock
Gross Profit	Net Sales – COGS
Net Profit	GP – Indirect Exp + Other Incomes
Closing capital	Opening cap + Net profit – Drawings + Additional cap
SLM depreciation	(Cost – Salvage) / Useful life
WDV depreciation	(WDV at start) × rate
WDV rate	$r = 1 - (S/C)^{(1/n)}$
Closing stock value	Lower of Cost or NRV
Subscription due (NPO)	Received + closing OS – opening OS – closing adv + opening adv

A.2 Constitutional / institutional anchors

Body	Article	Tenure
CAG	148	6 yrs / 65 yrs
FC	280	every 5 yrs (1 + 4 members)
RBI	RBI Act 1934	Governor 3 yrs
EC	324	6 yrs / 65 yrs
NITI Aayog	Cabinet resolution Jan 2015	not statutory
Annual Financial Statement	112	annual
Money Bill	110	
Consolidated Fund	266(1)	
Public Account	266(2)	
Contingency Fund	267	₹30,000 crore
GST Council	279A	from 2016

A.3 Economic ratios and aggregates

Concept	Formula
GDP at MP	Final-goods value at market prices, within country
GDP at FC	GDP at MP – IT + Subsidies
GNP at MP	GDP at MP + NFIA
NDP at MP	GDP at MP – Depreciation
NNP at MP	GNP at MP – Depreciation
National Income (NNP at FC)	NNP at MP – IT + Subsidies
Per Capita Income	NI / Population
GDP deflator	Nominal GDP / Real GDP × 100
Multiplier	$1/(1-MPC) = 1/MPS$
Money Multiplier	1/CRR (simple model)

A.4 Deficits

Deficit	Formula
Revenue Deficit	Rev Exp – Rev Receipts

Deficit	Formula
Effective Revenue Deficit	RD – Grants for capital assets
Fiscal Deficit	Total Exp – (Rev Receipts + Non-debt Cap Receipts)
Primary Deficit	FD – Interest payments

A.5 Demand-Supply / Cost

Concept	Formula
Price elasticity (Ed)	$\% \Delta Q / \% \Delta P$
Income elasticity (Ey)	$\% \Delta Q / \% \Delta Y$
Cross elasticity (Exy)	$\% \Delta Q_x / \% \Delta P_y$
AC = TC/Q	TC = TFC + TVC
AFC = TFC/Q	falling rectangular hyperbola
AVC = TVC/Q	U-shaped
MC = $\Delta TC / \Delta Q$	U-shaped; cuts AC at minimum
Equilibrium	MR = MC (rising MC)
Under PC	AR = MR = P; LR profit = normal
Under monopoly	AR > MR; can earn supernormal profit
Two-good budget line	$P_x \cdot X + P_y \cdot Y = M$
Consumer equilibrium (ordinal)	$MRS_{xy} = P_x / P_y$

A.6 Money & banking key rates (illustrative)

Tool	Current value (illustrative)
Repo	6.50 %
Reverse repo	3.35 % (now via SDF)
SDF	6.25 %
MSF	6.75 %
Bank rate	6.75 %
CRR	4.50 %
SLR	18.00 %
Inflation target (CPI)	4 % $\pm$ 2 %

A.7 BoP — quick map

Account	Components
Current	Trade in goods + services + income + transfers
Capital	FDI + FPI + ECB + NRI deposits + banking capital + reserves

A.8 GST quick facts

Item	Detail
Date	1 July 2017
Constitutional Amendment	101st (2016)
Council article	279A
Tax types	CGST + SGST + IGST + UTGST
Slabs	0, 5, 12, 18, 28 (+ cess)
Voting	Centre 1/3, States 2/3; decisions need 3/4ths weighted majority

APPENDIX B — ULTIMATE DEPTH TABLES

B.1 All Finance Commissions of India

FC	Years (period)	Chairman	Devolution %	Notable
1st	1952–57	K.C. Neogy	—	First FC
2nd	1957–62	K. Santhanam	—	
3rd	1962–66	A.K. Chanda	—	
4th	1966–69	P.V. Rajamannar	—	
5th	1969–74	Mahavir Tyagi	—	
6th	1974–79	K. Brahmananda Reddy	—	
7th	1979–84	J.M. Shelat	—	
8th	1984–89	Y.V. Chavan	—	
9th	1989–95	N.K.P. Salve	—	
10th	1995–2000	K.C. Pant	29 %	
11th	2000–05	A.M. Khusro	29.5 %	
12th	2005–10	C. Rangarajan	30.5 %	
13th	2010–15	Vijay L. Kelkar	32 %	
14th	2015–20	Y.V. Reddy	42 %	Big jump
15th	2021–26	N.K. Singh	41 %	Population data: 2011 census
16th	2026–31	Arvind Panagariya	TBD	Recommendations submitted by Oct 2025

B.2 RBI Governors (recent)

#	Name	Tenure	Notable
23	C. Rangarajan	1992–97	Liberalisation era
24	Bimal Jalan	1997–2003	Asian crisis management
25	Y.V. Reddy	2003–08	Pre-crisis caution
26	D. Subbarao	2008–13	Post-2008 crisis
27	Raghuram Rajan	2013–16	AQR (Asset Quality Review), inflation targeting
28	Urjit Patel	2016–18	First MPC; demonetisation
29	Shaktikanta Das	2018–24	COVID response, longest-serving recently
30	Sanjay Malhotra	Dec 2024–	Current Governor

B.3 Major economic reforms / Acts

Year	Reform / Act
1934	RBI Act
1949	Banking Regulation Act; RBI nationalised
1956	Industrial Policy Resolution; LIC nationalised
1969	First bank nationalisation (14 banks)
1973	FERA
1980	Second bank nationalisation (6 banks)
1991	New Industrial Policy + New Economic Policy
1992	SEBI Act
1997	TRAI Act
1999	FEMA + IRDA
2002	Competition Act + Sarfaesi Act
2003	FRBM Act + Fiscal Responsibility & Budget Management

Year	Reform / Act
2005	RTI Act + MGNREGA
2013	Companies Act 2013 + DBT
2014	PMJDY
2016	GST Constitutional Amendment + Insolvency & Bankruptcy Code (IBC) + Aadhaar Act
2017	GST implemented (1 July)
2019	NSO formed (CSO + NSSO merger)
2022	Standing Deposit Facility (SDF)
2023	Digital Personal Data Protection Act

B.4 Major bank nationalisation milestones

Year	Banks
1955	Imperial Bank → State Bank of India
1969	14 commercial banks nationalised (deposits ≥ ₹50 crore)
1980	6 more banks nationalised
1993	New Bank of India merged with PNB
2017	SBI absorbed 5 associate banks + Bharatiya Mahila Bank
2019	Mega merger: 10 PSBs into 4
2024	12 PSBs (post-merger) + SBI = total 12 PSBs

B.5 Five-Year Plans (history quick reference)

Plan	Period	Focus	Key model
1st	1951–56	Agriculture	Harrod - Domar
2nd	1956–61	Heavy industry	Mahalanobis (P.C. Mahalanobis)
3rd	1961–66	Self - reliance	Sukhamoy Chakravarty
4th	1969–74	Growth + stability	
5th	1974–79	Garibi Hatao	D.P. Dhar
6th	1980–85	Modernisation	
7th	1985–90	Food, work, productivity	
8th	1992–97	HRD; era of LPG	
9th	1997–2002	Growth with social justice	
10th	2002–07	Doubling per capita income	
11th	2007–12	Inclusive growth	
12th	2012–17	Faster, sustainable, inclusive	(Last FYP)

*Planning Commission abolished 1 Jan 2015 → replaced by **NITI Aayog**. No FYPs since 12th.*

B.6 Major poverty / employment committees

Committee	Year	Subject
Dadabhai Naoroji	1901	First poverty estimate
Y.K. Alagh	1979	Poverty line via calorie norms
Lakdawala	1993	State-specific PLs
Tendulkar	2009	Used MPCE; poverty line ₹816/₹1000 (rural/urban) for 2011–12
Rangarajan	2014	Higher PL ₹972/₹1407
Saxena	2009	BPL identification
Hashim	2012	Urban poverty

Committee	Year	Subject
Lakdawala (NSC)	various	NSS surveys

B.7 Major International Financial Institutions

Institution	Headquarters	Year	Role
IMF	Washington DC	1944 (Bretton Woods)	Monetary stability, BoP support
World Bank (IBRD)	Washington DC	1944	Long-term development loans
IDA	Washington DC	1960	Soft loans to poorest countries
IFC	Washington DC	1956	Private-sector financing
ADB	Manila	1966	Asian development
WTO	Geneva	1995	Successor to GATT (1947)
BIS	Basel	1930	Central bankers' bank (Basel norms)
AIIB	Beijing	2016	Asian Infra
NDB (BRICS Bank)	Shanghai	2014	Infrastructure for BRICS countries

APPENDIX C — Plain-English Concept Primers (first-time readers)

C.1 What is "double-entry book-keeping"? (Ch 2 primer)

Every transaction has two sides: someone gives, someone receives. If you pay ₹100 to a vendor, your cash decreases by ₹100 and the vendor's account is settled by ₹100. **Double-entry** records both sides every time.

Why? Because it makes errors visible (totals must always equal), and because it shows clearly where money came from and where it went.

The **debit** side records what came in (assets, expenses); the **credit** side records what went out (liabilities, income, capital). This is mechanical — once you internalise the rule, every entry is automatic.

C.2 What is a "balance sheet"? (Ch 4 primer)

A balance sheet is a **snapshot** of a business at one moment in time. It says: here's what we own (assets), here's what we owe (liabilities), and here's the difference (capital — what belongs to the owner).

The fundamental identity is **Assets = Liabilities + Capital**. If you sold all the assets and paid off all the debts, the leftover would belong to the owner. That's capital.

The P&L (profit & loss) is different — it's a **flow** statement, showing what happened over a period (a year, a quarter). Profit from P&L is added to Capital on the balance sheet — that's how the two are linked.

C.3 What is "depreciation"? (Ch 5 primer)

A delivery van costing ₹6 lakh today won't be worth ₹6 lakh in 5 years — it loses value due to wear, age, technology change. Depreciation accounts for this fall.

The accountant **spreads the cost** over the asset's useful life so each year's profit reflects the asset's "true" usage cost. SLM (Straight Line) spreads equally; WDV (Written Down Value) charges more in early years.

Depreciation is a **non-cash expense** — no money leaves the business, but profit is reduced. This serves two purposes: (1) tells the truth about profitability, (2) reduces taxable income, helping save for replacement.

C.4 What is the "Finance Commission"? (Ch 7 primer)

India is a federal country: the Centre collects most taxes, but the States provide most public services. The Finance Commission (FC) is the **referee** that decides how the tax pool is split between the two.

Every 5 years, the President appoints a new FC (Article 280). The Commission studies population, area, fiscal needs, performance, etc., and recommends: - What % of central taxes go to States (vertical) - How that share is split among the 28 States (horizontal) - What grants-in-aid States deserve

The latest (15th FC under N.K. Singh) gave States 41 % of the divisible tax pool. The 16th FC (Arvind Panagariya) is preparing recommendations for 2026-31.

C.5 What is "elasticity"? (Ch 9 primer)

Elasticity measures **how responsive** demand (or supply) is to price changes.

If a 10 % price increase causes only a 2 % drop in demand, the good is **inelastic** ($E_d = 0.2 < 1$). People keep buying despite the price hike. Examples: salt, life-saving medicine, petrol in the short run.

If a 10 % price increase causes a 25 % drop in demand, the good is **elastic** ($E_d = 2.5 > 1$). People switch to alternatives. Examples: branded clothing, restaurant meals, holidays.

Elasticity matters for **business pricing** (raising price of an inelastic good increases revenue) and for **government taxation** (taxing inelastic goods raises more revenue without hurting demand much).

C.6 What does the "RBI" do? (Ch 14 primer)

The Reserve Bank of India (estd 1935) is India's **central bank** — the bank of banks and the banker to the government. It does five big things:

1. **Issues currency** (the rupee notes you carry).
2. **Manages monetary policy** — sets interest rates to keep inflation around 4 %.
3. **Banker to the government** (Centre + States) — handles their accounts and debt.
4. **Banker's bank** — holds the reserves of all commercial banks; lends to them in emergencies.
5. **Regulator** — issues licences, sets rules, supervises banks.

When the RBI **raises** the repo rate (the rate at which it lends to banks), banks raise their rates, loans become costlier, demand cools, inflation falls. When it **cuts** the repo rate, the opposite happens. This is monetary policy.

C.7 What is a "fiscal deficit"? (Ch 15 primer)

The government's annual income is mainly tax + non-tax revenue. Its expenditure is salaries, subsidies, defence, infrastructure, interest on past borrowing. When **expenditure > income**, the gap is the deficit.

The most-watched number: **Fiscal Deficit** = Total Expenditure – (Revenue Receipts + Non-debt Capital Receipts). This is exactly **how much the government borrows in a year**.

Why does it matter? Borrowing today = repaying tomorrow with interest. High fiscal deficit can cause inflation, raise interest rates, crowd out private investment, hurt the rupee. The FRBM Act (2003) tries to keep it under 3 % of GDP (currently relaxed to ~5 % post-COVID).

C.8 What is "GST"? (Ch 15 primer)

Before GST (pre-2017), India had a maze of indirect taxes: excise (centre), service tax (centre), VAT (states), CST (inter-state), entry tax, octroi, luxury tax, etc. Each had its own rate, paperwork, and compliance burden.

GST (1 July 2017) merged all these into a single tax on the supply of goods and services at the **point of consumption**. It is collected in 3 forms: CGST (Centre), SGST (State), IGST (inter-state). Slabs: 0, 5, 12, 18, 28 % + cess on demerit goods.

Benefits: simpler compliance, less cascading of taxes, "one nation, one market". Challenges: small businesses' compliance burden, multiple slabs, exclusion of petroleum/alcohol from GST.

APPENDIX D — Full-length Mock Test (100 Qs, with answer key)

Instructions. 120 minutes. 2 marks per Q. -0.5 per wrong. Calculator allowed.

Section I — Finance & Accounts (Qs 1-40)

1. Accounting equation states: Assets = ?
2. Accounting standards in India are issued by?
3. Principle of "anticipate no profit, provide for all losses" is called?
4. Recording assets at historical cost is which concept?
5. Going concern means?
6. ICAI was established in?
7. Salary outstanding (not paid) is which kind of account?
8. Cash purchases of ₹2000 — debit account?
9. Cash book balance: ₹5000; cheques issued not presented ₹1200; cheques deposited not cleared ₹800. Pass book balance?
10. Days of grace on a bill?
11. Bill drawn 1 March for 2 months — maturity?
12. Drawer of a bill is also called?
13. Suspense A/c is used to balance?
14. Trial balance verifies?
15. Sales returns book records?
16. Direct expenses go in?
17. Closing stock valued at?
18. Net profit transferred to?
19. Outstanding wages in B/S?
20. Goodwill is?
21. Carriage inward goes in?
22. Carriage outward goes in?
23. SLM annual depreciation = ?
24. Method allowed under Indian Income Tax for depreciation?
25. Conservatism applied to inventory means stock valued at?
26. Sale of fixed asset is which receipt?
27. Heavy advertisement spread over years is?
28. AS-2 prohibits which inventory method?
29. FIFO in rising prices → profit is?
30. Plant cost ₹100,000, salvage ₹10,000, life 9 yrs. SLM annual depreciation?
31. NPO uses ... instead of P&L?
32. NPO uses ... instead of capital?
33. Receipts & Payments A/c records?
34. Surplus = Excess of ...?
35. Self-balancing system has how many ledgers?
36. Subscription due (current yr) formula?
37. Donations for building → which fund?
38. Personal accounts include?
39. Drawings appears on which side of capital A/c?
40. Bank overdraft is a?

Section II — Constitutional + Macro Economics (Qs 41-65)

41. Article for CAG?
42. Tenure of CAG?
43. CAG appointed by?
44. "Guardian of the public purse" refers to?
45. Article for Finance Commission?
46. FC composition?
47. Devolution % under 14th FC?
48. Chairman of 15th FC?

- 49. Chairman of 16th FC?
- 50. Article for budget?
- 51. Article for Money Bill?
- 52. Three funds in Constitution?
- 53. Article for Contingency Fund?
- 54. Father of economics?
- 55. Scarcity definition by?
- 56. Welfare definition by?
- 57. Coined micro/macro terms?
- 58. PPC shape?
- 59. Outside PPC = ?
- 60. Largest sector in India's GDP?
- 61. Largest sector in India's employment?
- 62. Tendulkar Committee subject?
- 63. Body for measuring NI in India?
- 64. Census 2011 population?
- 65. Demographic dividend window?

Section III — Money & Banking, BoP, Markets (Qs 66-85)

- 66. Law of demand: $P \uparrow \rightarrow Q?$
- 67. $E_d = 0$ means?
- 68. Cross elasticity of complements?
- 69. Income elasticity of inferior good?
- 70. IC convex to origin because?
- 71. Slope of budget line?
- 72. Veblen good is exception to?
- 73. Number of sellers in PC?
- 74. Demand curve shape in PC?
- 75. AR vs MR in monopoly?
- 76. Long-run profit in monopolistic competition?
- 77. Kinked demand by?
- 78. Mutual interdependence is characteristic of?
- 79. RBI established?
- 80. RBI nationalised?
- 81. First Governor?
- 82. Sole currency issuer?
- 83. $M3 = ?$
- 84. Inflation target of MPC?
- 85. MPC members?

Section IV — Fiscal, GST, Reforms, IT (Qs 86-100)

- 86. Fiscal Deficit formula?
- 87. Primary Deficit formula?
- 88. FRBM Act year?
- 89. GST CA Amendment number?
- 90. GST date?
- 91. GST Council article?
- 92. India joined WTO on?
- 93. FEMA replaced FERA in?
- 94. LPG full form?
- 95. NITI Aayog year?
- 96. Digital India launched on?
- 97. UMANG full form?
- 98. JAM trinity?
- 99. IT Act year?
- 100. Section 66A struck down in?

Answer key

#	Ans	#	Ans	#	Ans	#	Ans
1	Liabilities + Capital	26	Capital	51	110	76	Normal
2	ICAI	27	Deferred revenue exp	52	CFI, PA, CF	77	Sweezy
3	Conservatism	28	LIFO	53	267	78	Oligopoly
4	Cost concept	29	Higher	54	Adam Smith	79	1 Apr 1935
5	Business continues	30	₹10,000	55	Robbins	80	1 Jan 1949
6	1949	31	Income & Exp A/c	56	Marshall	81	Osborne Smith
7	Liability (representative personal)	32	Capital Fund	57	Frisch	82	RBI
8	Purchases A/c	33	All cash transactions of all years	58	Concave to origin	83	M1 + Time deposits
9	₹5,400	34	Income over Expenditure	59	Infeasible	84	4 % ± 2 %
10	3 days	35	3	60	Services (~55 %)	85	6
11	4 May	36	Received + closing OS – opening OS – closing adv + opening adv	61	Agriculture (~45 %)	86	TE – (RR + Non-debt CR)
12	Maker	37	Capital Fund	62	Poverty estimation	87	FD – Interest
13	Trial Balance	38	Persons, firms, banks	63	NSO	88	2003
14	Arithmetical accuracy	39	Debit (deduction)	64	~121 crore	89	101st
15	Goods returned by customers	40	Liability	65	2018-2050	90	1 July 2017
16	Trading A/c	41	148	66	Falls	91	279A
17	Lower of cost or NRV	42	6 yrs / 65 yrs	67	Perfectly inelastic	92	1 Jan 1995
18	Capital A/c	43	President	68	Negative	93	1999
19	Current liability	44	CAG	69	Negative	94	Liberalisation, Privatisation, Globalisation
20	Intangible fixed asset	45	280	70	Diminishing MRS	95	2015
21	Trading A/c (Dr)	46	1 + 4	71	–Px/Py	96	1 July 2015
22	P&L A/c (Dr)	47	42 %	72	Law of demand	97	Unified Mobile App for New-age Governance
23	(Cost – Salvage)/Life	48	N.K. Singh	73	Very many	98	Jan Dhan + Aadhaar + Mobile
24	WDV (block of assets)	49	Arvind Panagariya	74	Horizontal	99	2000
25	Lower of cost or NRV	50	112	75	AR > MR	100	2015 (Shreya Singhal)

APPENDIX E — 7-Day Final Revision Plan

Day	Morning (3 h)	Evening (3 h)	Recall pass
D-7	Ch 1 + Ch 2 (re-read theory + mini-mocks)	Ch 3 (BRS + Bills) — solve all worked examples	Closed-book recall of Ch 1-3
D-6	Ch 4 + Ch 5 (final accounts + depreciation)	Ch 6 (NPO)	Recall Ch 4-6
D-5	Ch 7 (CAG + FC) + Ch 8 (basic econ)	Ch 9 (Demand & Supply)	Recall Ch 7-9
D-4	Ch 10 (Production & Cost) + Ch 11 (Markets)	Ch 12 (Indian Economy) — heavy facts	Recall Ch 10-12
D-3	Ch 13 (Reforms) + Ch 14 (Money & Banking)	Ch 15 (Fiscal/Budget/BoP/FRBM)	Recall Ch 13-15
D-2	Ch 16 (IT in Governance) + ULTIMATE TABLES (FCs, RBI Gvs, Acts)	Master Formula Sheet — write from memory twice	All recall prompts in one sitting
D-1	Full-length mock (Appendix D) under timed conditions	Score honestly. Re-read chapters where you erred.	Light read of trap-recognition cards
D-0 (test day)	Calm 30-min glance at Master Formula Sheet	Test!	—

Test-day strategy.

- First sweep (40 min)** — knock out direct definition / fact / formula Qs (CAG article, FC chairman, GST date, FD formula). Skip anything > 90 sec.
- Second sweep (50 min)** — return to medium Qs (BRS, Final A/c adjustments, elasticity numericals, IS-LM concepts).
- Final sweep (25 min)** — attempt the leftover hard ones; mark unsure ones.
- Last 5 min** — recheck answer-button selections.
- Negative-marking rule.** Eliminate at least 2 of 4 options before attempting. If you cannot eliminate any, leave it.

APPENDIX F — Common errors that cost AAO candidates the 95 % mark

#	Error	Why it happens	Correct fix
1	"GAAP issued by RBI"	Confusion with monetary regulator	ICAI issues; MCA notifies
2	Sales book includes cash sales	Lazy memory	Cash sales → Cash book; Sales book = credit sales only
3	LIFO valid in India	Following US GAAP	Prohibited by AS-2; only FIFO and Weighted Avg
4	Closing stock at cost always	Forgot conservatism	Lower of cost or NRV
5	Capital expenditure in P&L	Mismatched concept	Capital → Balance Sheet (asset); Revenue → P&L
6	Drawings = expense	Confusion	Drawings = reduction from capital, not expense
7	CAG performs Comptroller function	Misread	India's CAG is only Auditor (no comptroller function unlike UK)
8	FC has 5 + 5 members	Misread	1 Chairman + 4 = 5 total
9	14th FC devolution = 32 %	Confused with 13th	14th = 42 %; 13th = 32 %; 15th = 41 %
10	RBI established 1949	Confusion	Established 1935; nationalised 1949
11	Bank rate is collateralised	Confused with repo	Bank rate uncollateralised; repo against govt securities
12	MPC has 12 members	Misread	6 (3 RBI + 3 govt)
13	CRR held in govt securities	Confused with SLR	CRR = cash with RBI; SLR = govt securities + gold
14	Inflation target = 6 %	Misread	Target 4 %, ±2 % tolerance band
15	NI = GDP at MP	Concept slip	NI = NNP at FC
16	Tendulkar = 2014	Confused with Rangarajan	Tendulkar 2009; Rangarajan 2014
17	Largest sector in GDP = Agriculture	Reverse	Services biggest in GDP; Agriculture biggest in employment
18	Annual Financial Statement under Art. 110	Confused with Money Bill	Art. 112 (AFS); Art. 110 (Money Bill)
19	Fiscal deficit excludes interest	Misread	FD includes interest; Primary deficit excludes
20	GST has 3 slabs	Misread	5 slabs (0/5/12/18/28) + cess
21	LPG = Liquefied Petroleum Gas	Wrong context	In economics: Liberalisation, Privatisation, Globalisation
22	FERA replaced FEMA	Reverse	FEMA (1999) replaced FERA (1973)
23	UIDAI Act 2009	Confusion	UIDAI estd 2009; Aadhaar Act 2016
24	Section 66A still valid	Outdated	Struck down 2015 (Shreya Singhal)
25	Producer operates in Stage I (production)	Wrong	Stage II (rational producer)

APPENDIX G — Memory tricks (mnemonics that stick)

Concept	Mnemonic
Debit / Credit (modern)	" DEAD CLIC " — Debit: Expenses + Assets + Drawings; Credit: Liabilities + Income + Capital
Order of accounting cycle	" Record → Classify → Summarise → Communicate " (RCSC)
Errors not caught by Trial Balance	" O mission, C ommission, P rinciple, C ompensation, O riginal entry" — OCPCO
BRS direction	"Add what makes pass book bigger; subtract what makes it smaller" (relative to cash book)
Bill of exchange parties	D rawer (creator) → D rawee (debtor) → P ayee (recipient); 3 D-P chain
Depreciation methods	" SLM is S imple straight; WDV W inds down"
Inventory valuation in India	"AS-2 says FIFO yes, LIFO no"
CAG = Auditor only	"India's CAG is only A , not C omptroller"
FC composition	"1 Chair + 4 = 5 hands decide tax sharing every 5 years"
RBI dates	" 35-49 : Born and naturalised in 14 years (1935 → 1949)"
MPC structure	" 3+3=6 : half RBI, half govt; Governor casts the tie"
Money supply order	"M0 → M1 → M2 → M3 → M4 — bigger number, broader money, lower liquidity"
Inflation target	" 4±2 : aim for 4 %, never breach 6 % upper"
LPG	"Let People Globalise"
JAM trinity	"Jan Dhan + Aadhaar + Mobile = JAM " (sticky financial inclusion)
FRBM target	" 3% deficit, 60% debt-to-GDP, 40% Centre + 20% States — 3-60-40-20"
Articles to remember	CAG = 148, FC = 280, Budget = 112, Money Bill = 110, GST Council = 279A, EC = 324
GST	" G oods and S ervices T ax — 1 nation, 1 tax , 1 July 2017"

Final words

This book is your **complete** AAO Paper III reference: 16 chapters covering the entire Finance & Accounts + Economics & Governance syllabus, with examiner-mindset analysis, depth tables, formula sheet, full-length mock, primers and revision plan. Re-read twice, drill the worked examples, time yourself on the mock — and the AAO paper becomes a routine.

Read smart. Drill hard. Sleep well. **190+/200 is yours.**

— Pariksha365